INTELLIGENT ATTENDANCE CONTROL AND STUDENT PROGRESS MONITORING PLATFORM – EASWARI ERP PORTAL

PROJECT REPORT

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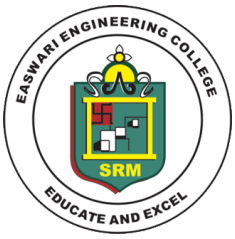
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EASWARI ENGINEERING COLLEGE, CHENNAI
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BONAFIDE CERTIFICATE

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PROJECT REPORT

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ABSTRACT

In most colleges, attendance and internal marks are still kept in paper registers and spreadsheets that different people update at different times. The figures seen by a student, a class coordinator and the examination cell therefore often disagree, a class can be marked by a teacher who does not handle it, and short attendance is discovered too late to be corrected. This project presents the Easwari ERP Portal, a web platform for attendance control and student progress monitoring that keeps one authoritative record of each fact and controls who may change it. Student, Faculty and Administrator portals sit behind a single sign-in. New accounts receive a random temporary password that must be changed at the first login, and students may create a six-digit quick code that is locked after five wrong attempts.

Attendance is stored once for each student, subject offering, date and period under a database constraint that makes double marking impossible; the period, day and subject views are all derived from that record. The class coordinator may mark and correct every period of the class, a subject teacher may change only his or her own subject, and the administrator oversees every class, with the server enforcing these rules for every request. Internal marks are visible to students only after publication. Students see attendance, 75% eligibility, marks, practicals, university results, fees, leave, placement matches and a Career Hub. A separate FastAPI service computes an explainable academic-risk score.

The system uses Java 17 and Spring Boot 3.2.5, React 18 with Vite, JSON Web Tokens with BCrypt, an embedded H2 database that moves to PostgreSQL by configuration, and a Python scoring service. A trial institution of five departments, ten classes, 59 students and 25 staff members was seeded and exercised. In 68 recorded test cases and 11 unit tests the rules held in all but one case; the single defect found is reported with its fix.

KEYWORDS: Attendance Management, Student Progress Monitoring, College ERP, Role-Based Access Control, Spring Boot, React, JSON Web Token, Academic Risk Score, FastAPI

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE
1.	**INTRODUCTION**	**1**
	1.1 GENERAL	1
	1.2 OBJECTIVE	1
	1.3 PROBLEM DESCRIPTION	2
	1.4 OBJECTIVES OF THE PROJECT	2
	1.5 SCOPE OF THE PROJECT	3
	1.6 ORGANISATION OF THE PROJECT REPORT	3
	1.7 SUMMARY	4
2.	**LITERATURE SURVEY**	**5**
	2.1 INTRODUCTION	5
	2.2 EXISTING SYSTEM	5
	2.3 LIMITATIONS OF EXISTING SYSTEM	5
	2.4 RELATED WORKS AND OBSERVATIONS	6
	2.5 PROPOSED SYSTEM OVERVIEW	7
	2.6 SUMMARY	8
3.	**SYSTEM DESIGN**	**9**
	3.1 INTRODUCTION	9
	3.2 PROPOSED SYSTEM	9
	3.3 FEASIBILITY STUDY	10
	3.4 SYSTEM ARCHITECTURE	10
	3.4.1 AUTHENTICATION AND SESSION	11
	3.4.2 ATTENDANCE AND AUTHORITY	12
	3.4.3 DATA MODEL	12
	3.4.4 RISK SCORE	13
	3.5 SUMMARY	14
4.	**METHODOLOGY**	**15**
	4.1 INTRODUCTION	15
	4.2 HARDWARE AND SOFTWARE SPECIFICATIONS	15
	4.3 MODULE IDENTIFICATION	17
	4.3.1 AUTHENTICATION AND ACCOUNT MODULE	17
	4.3.2 ATTENDANCE MODULE	17
	4.3.3 MARKS AND EXAMINATION MODULE	17
	4.3.4 FEES, SERVICES AND LEAVE MODULE	18

4.3.5 ADMINISTRATION AND AUDIT MODULE 18
4.3.6 INTELLIGENCE AND CAREER HUB MODULE 18
4.4 ALGORITHM / PSEUDOCODE 18
4.5 TOOLS USED 19
4.5.1 DATASET 19
4.5.2 IMPLEMENTATION DETAILS 19
4.6 SNAPSHOTS 20
4.7 SUMMARY 27
5. TESTING AND RESULTS 28
5.1 TESTING STRATEGIES 28
5.2 TEST CASES AND EXECUTION 28
5.3 RESULTS AND SCREENSHOTS 36
6. CONCLUSION 44
6.1 INTRODUCTION 44
6.2 CONCLUSION 44
6.3 SUMMARY AND FUTURE SCOPE 45
SAMPLE SOURCE CODE 47
REFERENCES 48

LIST OF TABLES

TABLE NO.	TITLE	PAGE NO.
2.4.1	COMPARISON OF RELATED WORK AND THE PROPOSED SYSTEM	7
4.1	HARDWARE REQUIREMENTS	15
4.2	SOFTWARE REQUIREMENTS	16
4.3	DIVISION OF WORK	17
4.4	SERVER ENDPOINTS	20
5.2.1	TEST CASES AND RESULTS	29
5.3.1	QUANTITATIVE RESULTS	36
5.3.2	RISK SCORE FOR THREE PROFILES	37

LIST OF FIGURES

FIGURE NO.	TITLE	PAGE NO.
3.1	USE CASE DIAGRAM	10
3.2	SYSTEM ARCHITECTURE	11
3.3	SIGN-IN AND FIRST-LOGIN FLOW	12
3.4	AUTHORITY CHECK FOR EVERY ATTENDANCE REQUEST	12
3.5	CORE DATA MODEL CENTRED ON THE SUBJECT OFFERING	13
3.6	RISK SCORE COMPUTATION	13
4.7	WELCOME AND ROLE SELECTION	21
4.8	STUDENT SIGN-IN	21
4.9	QUICK-CODE SIGN-IN	22
4.10	ADMINISTRATOR DASHBOARD	22
4.11	ACCESS MANAGEMENT	23
4.12	CREATE AN ACCOUNT	23
4.13	DEPARTMENTS AND CLASSES	24
4.14	COURSES AND TIMETABLE	24
4.15	FACULTY DASHBOARD	25
4.16	ATTENDANCE ENTRY BY THE CLASS COORDINATOR	25
4.17	ATTENDANCE ENTRY BY A SUBJECT TEACHER	26
4.18	MARKS ENTRY	26
5.1	STUDENT DASHBOARD	38
5.2	ATTENDANCE REGISTER	38
5.3	INTERNAL MARKS	39

FIGURE NO.	TITLE	PAGE NO.
5.4	UNIVERSITY EXAMINATION RESULTS	39
5.5	FEE PORTAL	40
5.6	LEAVE AND ON-DUTY APPLICATION	40
5.7	RISK AND ALERTS	41
5.8	LEAVE REQUESTS (ADMINISTRATION)	41
5.9	LEAVE REVIEW	42
5.10	AUDIT TRAIL	42
5.11	PLACEMENT	43
5.12	CAREER HUB OVERVIEW	43

GLOSSARY OF TECHNICAL TERMS

TERM	MEANING
API	Application Programming Interface
BCrypt	Blowfish-based password hashing function
CGPA	Cumulative Grade Point Average
CORS	Cross-Origin Resource Sharing
CSS	Cascading Style Sheets
ERP	Enterprise Resource Planning
HOD	Head of the Department
HTTP	Hyper Text Transfer Protocol
IA	Internal Assessment
JPA	Java Persistence API
JSON	JavaScript Object Notation
JUnit	Java unit testing framework
JWT	JSON Web Token
OD	On Duty
ORM	Object Relational Mapping
PDF	Portable Document Format
REST	Representational State Transfer
SQL	Structured Query Language
SVG	Scalable Vector Graphics
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 GENERAL

Attendance and internal assessment are the two records on which almost every academic decision in an engineering college depends. A student is allowed into the end-semester examination only if the attendance requirement of 75% is met, and the internal marks contribute directly to the final grade. Yet in many colleges these records are still written in paper registers and typed into spreadsheets by different people at different times.

The result is a familiar set of problems. The attendance shown on the notice board does not match the figure held by the class coordinator, a period may be marked twice or not at all, a teacher can alter the attendance of a class that he or she does not handle, and a student learns of short attendance only when the eligibility list is displayed, which is too late to correct it.

This project, the Easwari ERP Portal, is a web platform that places attendance control, internal assessment, examination results, fee collection, campus service requests, leave and on-duty approvals and placement preparation into one system. It is built around three role-specific portals for students, faculty and administrators, behind a single secure sign-in.

1.2 OBJECTIVE

The objective of the project is to design and implement a modular, role-based web platform that keeps one authoritative record of attendance and internal assessment, enforces who may change which part of it, and turns the stored data into timely information for students, faculty and administrators.

1.3 PROBLEM DESCRIPTION

Attendance, internal marks, fees and student requests are kept in separate places and are updated by different people. Four weaknesses follow from this arrangement.

1. **No single record:** the same student can show different attendance on different lists because each list is a separate copy.
2. **No control over authority:** a paper register or a shared spreadsheet cannot stop a teacher from editing a class that he or she does not teach, and corrections are made by overwriting, so no one can tell who changed what.
3. **Late warning:** short attendance and falling internal marks are noticed at the end of the semester, when nothing can be done about them.
4. **Scattered services:** fee payment, leave, certificates and placement information are handled at different counters and on different notice boards.

A system is therefore needed in which each fact is stored once, every change is checked against the authority of the person making it, and the right view is shown to the right person.

1.4 OBJECTIVES OF THE PROJECT

- Provide Student, Faculty and Administrator portals behind one sign-in, with random temporary passwords for new accounts and a compulsory change at the first login.
- Record attendance per period under a database constraint that makes double marking impossible, and offer a single-period path and a whole-day path for faculty.
- Let the class coordinator mark and correct every period of the class, restrict a subject teacher to his or her own subject, and give the administrator oversight of every class.

- Capture internal assessments and laboratory components with a maximum mark for each assessment and release them to students only when published.
- Show every student the attendance register, subject-wise eligibility, internal marks, practicals, university results and CGPA.
- Provide a fee portal with simulated payment and receipts, a campus-services queue, and leave and on-duty applications with proof upload.
- Offer a Career Hub and placement module, and compute an explainable academic-risk score in a separate scoring service.

1.5 SCOPE OF THE PROJECT

The scope covers the React front end, the Spring Boot REST API, the relational data model, the FastAPI scoring service and the seeding of a complete trial institution consisting of five departments, ten classes, 59 students and 25 staff members. It includes authentication, role-based authorization, attendance, marks, university results, fees, campus services, leave and on-duty, placement, internships, the Career Hub and the audit trail.

The scope does not include a live payment gateway, biometric or RFID attendance capture, or integration with the University’s own examination system. Payment is simulated inside the application, and the data shown in this report is a seeded trial data set and not the college’s own records.

1.6 ORGANISATION OF THE PROJECT REPORT

Chapter 1 introduces the project, its objectives and its scope. Chapter 2 describes the existing system, its limitations and the related published work. Chapter 3 presents the proposed system, the feasibility study and the system architecture. Chapter 4 describes the methodology: the hardware and software, the division of work, the modules, the algorithms, the tools and snapshots of the running application. Chapter 5 presents the testing strategy,

the 68 recorded test cases and the results. Chapter 6 gives the conclusion and the future scope. The sample source code and the references follow.

1.7 SUMMARY

This chapter explained why attendance and internal marks need a single controlled record, defined the problem, and stated the objectives and the scope of the Easwari ERP Portal. The next chapter reviews the existing system and the related work.

CHAPTER 2
LITERATURE SURVEY

2.1 INTRODUCTION

This chapter describes how attendance and marks are handled today, lists the limitations of that practice, and reviews three published studies that are close to the problem: one on web-based attendance capture and two on early identification of students who are at risk. It ends with an overview of the proposed system.

2.2 EXISTING SYSTEM

In the existing manual or semi-digital process each faculty member marks attendance in a register at the end of the hour, the class coordinator consolidates the registers at the end of the month, and the figures are typed into a spreadsheet for display. Internal marks follow the same route: they are written on mark sheets, entered into a spreadsheet and later forwarded to the examination cell. Fees are paid at a counter and leave letters are submitted on paper.

Some colleges use a commercial campus package, but these are usually licensed per student, are difficult to adapt to the rules of an affiliated autonomous college, and rarely expose the rule about who may modify a record in a way that the college can inspect.

2.3 LIMITATIONS OF EXISTING SYSTEM

- The same fact is stored in several copies that drift apart.
- Nothing prevents a period from being marked twice, or a record from being changed by a person who has no authority over the class.
- Corrections overwrite the earlier value and leave no audit trail.

- Students cannot see their own attendance, marks and eligibility until a list is published.
- Warning signs such as falling attendance or a weak internal-assessment trend are not turned into an early alert.
- Fees, leave and service requests need physical visits and paper.

2.4 RELATED WORKS AND OBSERVATIONS

(a) QR code-based attendance. Ramadhani and Satria (2025) describe a web-based system in which each student has a unique QR code that is scanned with a camera-equipped device and saved to a central database, with login, report generation and automatic recording. The strength of this work is the removal of the paper register and the immediate availability of reports. It does not describe which member of staff may change which class, how a wrong mark is corrected, or how attendance relates to examination eligibility [1].

(b) Early warning by learning analytics. Akçapınar, Altun and Aşkar (2019) analysed the online-learning interaction data of 76 second-year students and compared classification algorithms to predict end-of-term performance early in the semester. Their result shows that early intervention is feasible when data is collected continuously, but the models depend on interaction logs from an online platform and were trained on a single course [2].

(c) Comparison of prediction methods. Howard, Meehan and Parnell (2017) compared eight prediction methods for the final grades of a large first-year statistics course and found that continuous-assessment marks were the most important predictor. This supports the weight given to internal marks in the risk score of the present system, and it also reminds the reader that such tools support decisions and do not replace them [3].

TABLE 2.4.1 COMPARISON OF RELATED WORK AND THE PROPOSED SYSTEM

Feature	QR attendance (2025)	Early warning (2019, 2017)	Easwari ERP Portal
Attendance capture	QR scan to central database	Not used	Period-level marking with unique constraint
Who may change a record	Not described	Not applicable	Coordinator: whole class; teacher: own subject; administrator: oversight
Internal marks	Not covered	Used as a predictor	Assessments with maximum marks and a publish gate
Early identification	Reports only	Classifiers on platform logs	Transparent weighted score with listed factors
Student self-service	Report view	Not described	Attendance, marks, fees, leave, services, placement

Observation. The attendance system captures attendance but does not define the authority to change it. The early-warning studies assume a learning-management platform that most affiliated engineering colleges do not run. What a college needs is an integrated record of attendance, marks, leave, fees and services that states clearly who may modify each part and that produces an early signal from data the college already holds.

2.5 PROPOSED SYSTEM OVERVIEW

The Easwari ERP Portal stores each attendance mark as one row identified by student, subject offering, date and period, and derives every report from those rows. Access rules are enforced on the server for every request, so hiding a menu item in the browser is never the only protection. Marks stay private to the teacher until they are published. A separate scoring service turns attendance, CGPA, internal-mark trajectory and eligibility into an

explainable risk score so that mentors can act early. The remaining modules bring fees, leave, services and placement into the same sign-in.

2.6 SUMMARY

The review shows a clear move from paper registers to central databases and from end-of-term results to early warning. The proposed system combines both directions for an affiliated engineering college and adds server-enforced authority rules, which the following chapters design, build and test.

CHAPTER 3
SYSTEM DESIGN

3.1 INTRODUCTION

This chapter presents the proposed system, its feasibility, and the architecture of the Easwari ERP Portal. The design rests on three principles: the Java service owns the data and the Python service owns the scoring models; security is layered; and attendance is stored once at the finest grain.

3.2 PROPOSED SYSTEM

The proposed system is a responsive web application with a component-based React front end, a Spring Boot REST API and a relational database. There are three kinds of account. An administrator provisions departments, programmes, classes, subject offerings, timetables and user accounts. A faculty member sees the classes he or she teaches and, if also the class coordinator, the whole class. A student sees only his or her own record.

The central data object is the subject offering, which is one subject taught to one class section by one member of staff. Timetable slots, attendance records and assessments hang from the offering, so a statement such as “Computer Networks attendance” has a precise meaning. Figure 3.1 shows the use cases of the three actors.

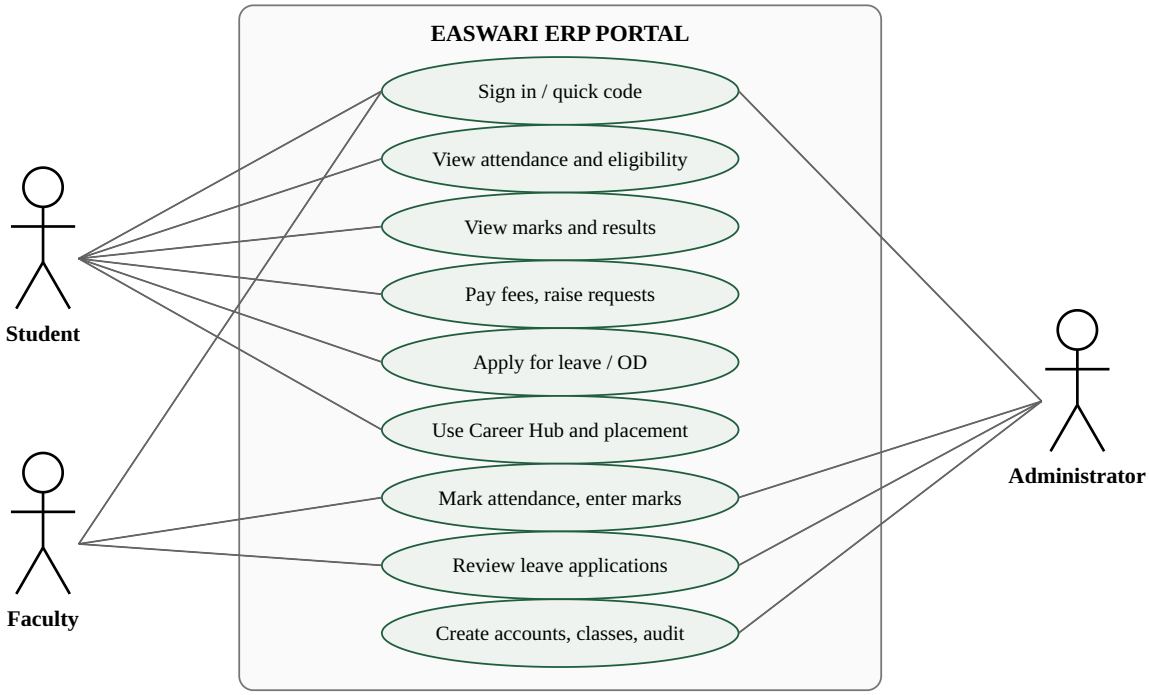


FIGURE 3.1 USE CASE DIAGRAM

3.3 FEASIBILITY STUDY

Technical feasibility. Every component is open source, runs on one laptop and was assembled into a working system during the project. **Economic feasibility.** No licence, hosting or hardware cost is needed for the demonstration, and a college could host the system on one small server. **Operational feasibility.** It depends on faculty marking attendance in the portal; the single-period path pre-fills the roster as present so that only the absentees are changed, which keeps the time per period to seconds. **Legal and data feasibility.** Passwords are hashed with BCrypt, temporary passwords are random, and a student can reach only his or her own record.

3.4 SYSTEM ARCHITECTURE

Figure 3.2 shows the layered architecture. The browser talks to the application layer through JSON over HTTP with a bearer token. The application layer reaches the relational database through Spring Data and the

scoring service through an HTTP client. The scoring service holds no database connection, so the portals continue to work when it is stopped.

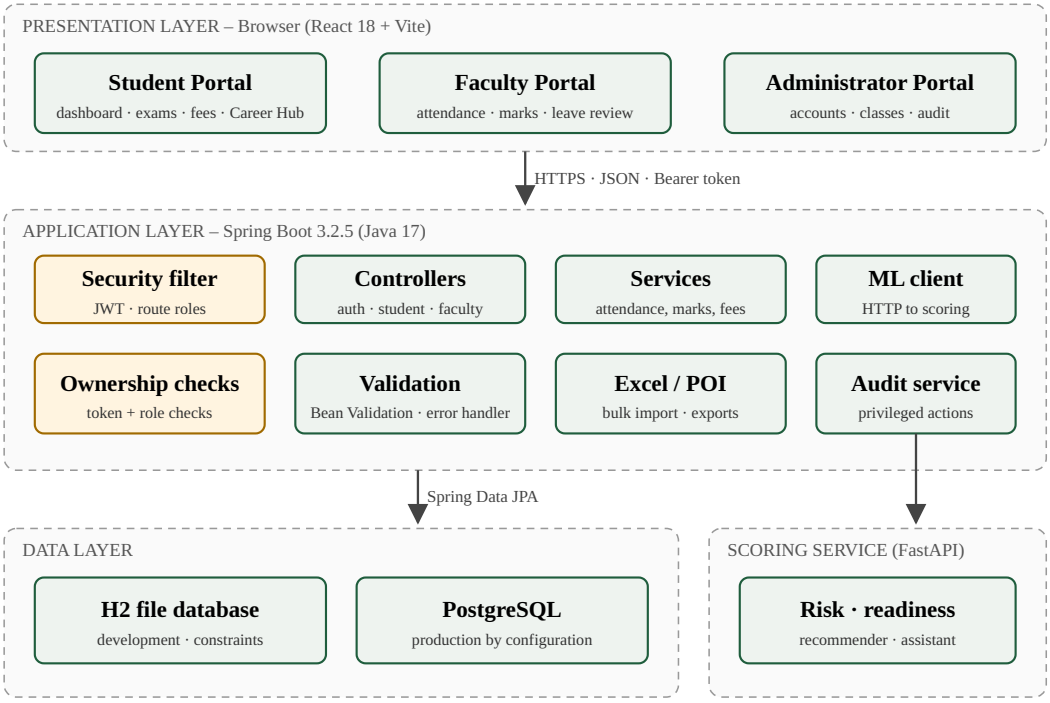


FIGURE 3.2 SYSTEM ARCHITECTURE

3.4.1 AUTHENTICATION AND SESSION

A single sign-in endpoint accepts a username and password and returns a signed token, the role, the display name and a flag that says whether the password must be changed. Accounts are created by the administrator with a random ten-character temporary password. A student can later create a six-digit quick code, which is stored hashed, works for students only and is locked for 15 minutes after five wrong attempts. The token is kept in the browser’s memory only, so a shared laboratory machine does not retain the session. Figure 3.3 shows the sign-in flow.

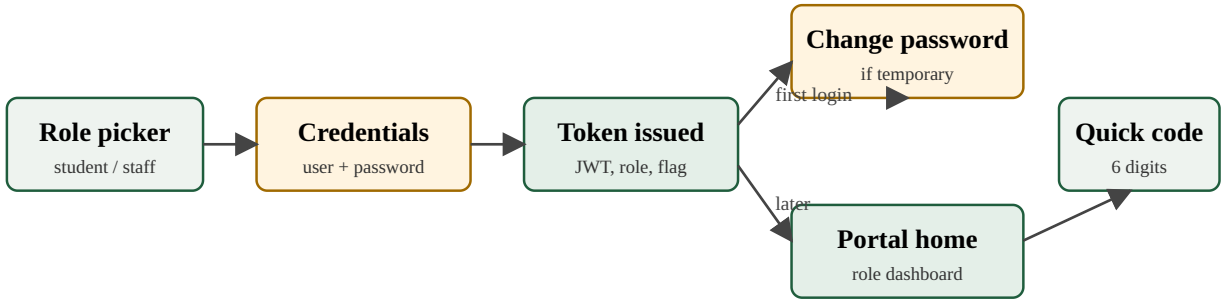


FIGURE 3.3 SIGN-IN AND FIRST-LOGIN FLOW

3.4.2 ATTENDANCE AND AUTHORITY

Attendance is stored once for each student, subject offering, date and period, under the unique constraint uk_attendance_slot, so marking a period again updates the existing row. The period view, the day view and the subject view are aggregations of the same rows and cannot disagree. Before any row is written, the server checks that the person is the teacher of the offering or the coordinator of the class, and that the students belong to that class (Figure 3.4).

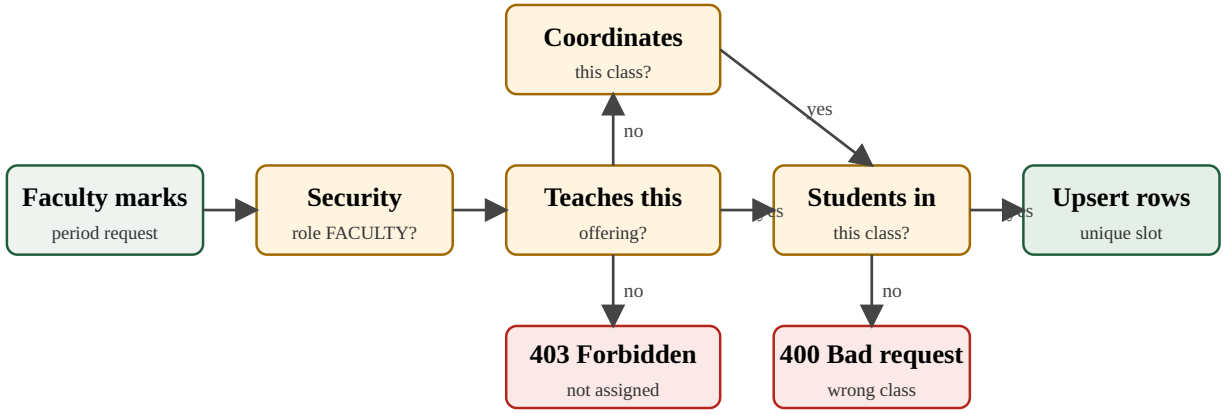


FIGURE 3.4 AUTHORITY CHECK FOR EVERY ATTENDANCE REQUEST

3.4.3 DATA MODEL

The data model in Figure 3.5 is organised around the subject offering. Attendance records and assessments belong to an offering and not directly

to a subject, because attendance in a subject is meaningful only once the section and the teacher are known.

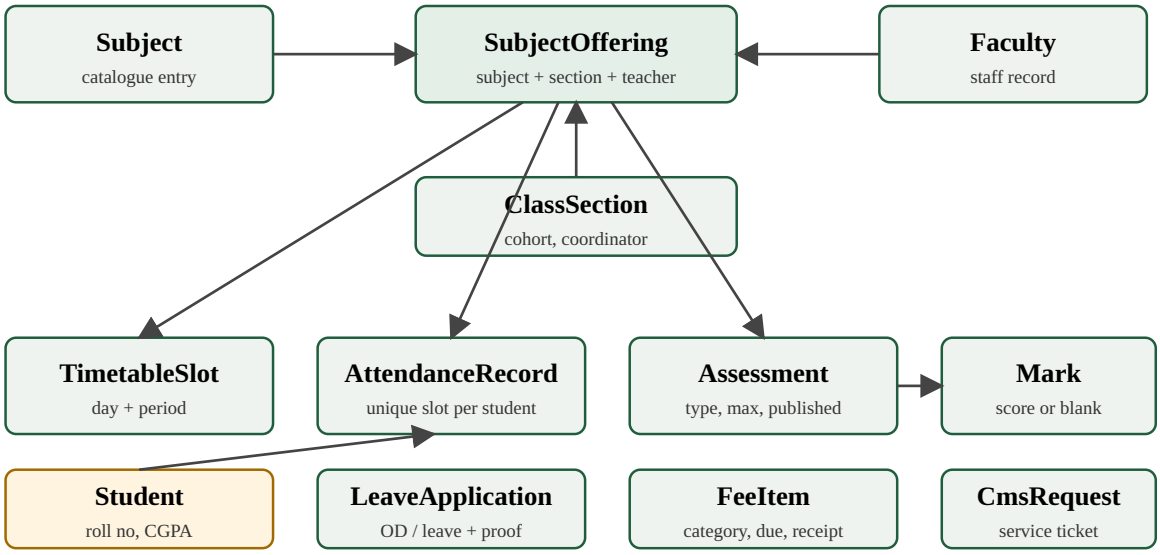


FIGURE 3.5 CORE DATA MODEL CENTRED ON THE SUBJECT OFFERING

3.4.4 RISK SCORE

The scoring service computes a weighted academic-risk score from attendance (0.35), CGPA (0.30), internal-mark trajectory (0.20) and eligibility (0.15). Scores below 35 are LOW, 35 to 59 MEDIUM and 60 or more HIGH. Each score lists the factors behind it, so the number can be explained to a student. Figure 3.6 shows the computation.

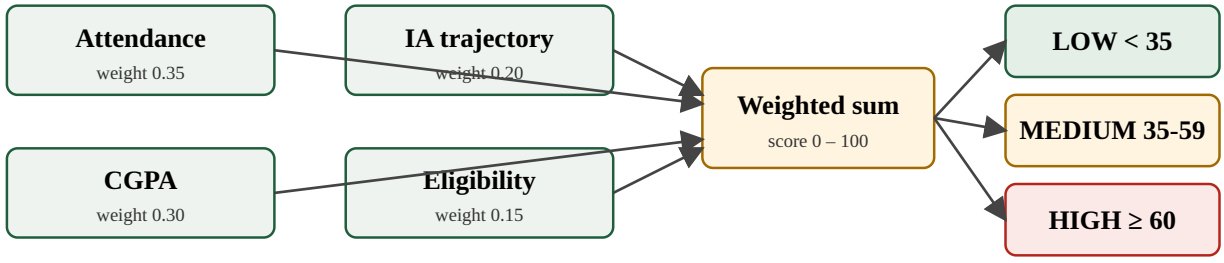


FIGURE 3.6 RISK SCORE COMPUTATION

3.5 SUMMARY

This chapter presented the proposed system, its feasibility, and the layered architecture with the sign-in flow, the attendance authority check, the data model and the risk score. The next chapter describes how the system was built.

CHAPTER 4
METHODOLOGY

4.1 INTRODUCTION

The methodology describes how the Easwari ERP Portal was built. The work was organised into six stages: (1) the domain model and database constraints, (2) authentication and security, (3) the attendance, marks and examination services, (4) the three portal front ends, (5) the fee, leave, service, placement and Career Hub modules with the scoring service, and (6) testing and evaluation.

The work was divided among the three team members by area. Table 4.3 lists the division of work.

4.2 HARDWARE AND SOFTWARE SPECIFICATIONS

TABLE 4.1 HARDWARE REQUIREMENTS

Component	Specification
Processor	Dual-core 2.0 GHz or higher (development: laptop CPU)
Memory (RAM)	4 GB minimum, 8 GB recommended
Storage	2 GB free space; the executable jar is about 72 MB
Display	1366 x 768 or higher
Network	Local network; wired or Wi-Fi campus network with HTTPS for deployment

TABLE 4.2 SOFTWARE REQUIREMENTS

Software	Version / Purpose
Operating system	macOS or Linux (development and test run); Windows also supported
Java and Spring Boot	Java 17 (jar also runs on Java 21), Spring Boot 3.2.5, Spring Data JPA, Spring Security, JJWT, Maven
Database	H2 in file mode; PostgreSQL by configuration profile
Front end	React 18, Vite 5, Recharts, Lucide icons, pdf.js, a WebGL background
Scoring service	Python 3 with FastAPI, Uvicorn and NumPy
Spreadsheet support	Apache POI 5.2.5 for bulk import and exports
Testing	JUnit 5 and Spring Security Test; Playwright-driven Chromium for the screenshots
Browser	Google Chrome or Microsoft Edge (latest)

JUSTIFICATION: Every component is open source and runs on an ordinary laptop, so no licence or server cost is needed. Spring Boot gives a dependable, well-supported back end with built-in security and validation, and H2 in file mode needs no installation yet enforces real relational constraints. React with Vite gives fast, component-based screens, and keeping the scoring models in a separate Python service lets each side be scaled or replaced independently.

4.3 MODULE IDENTIFICATION

TABLE 4.3 DIVISION OF WORK

Team member	Area	Responsibilities
REHAN J (310625104288)	Back end, database and integration	Domain model, constraints, authentication, attendance and marks services, administrator portal, scoring service, seeding and overall integration
SAKTHISARAN M (310625104313)	Front end	Sign-in and role picker, student portal screens, Career Hub, theme and responsive layout
SANJAY N (310620104322)	Testing and documentation	Test plan and the 68 recorded test cases, faculty-portal checks, screenshots and the project report

4.3.1 AUTHENTICATION AND ACCOUNT MODULE

Handles the password sign-in, the six-digit quick code for students, the forced change of a temporary password, and the provisioning of accounts by the administrator individually or from a spreadsheet. Passwords are hashed with BCrypt and tokens are signed JSON Web Tokens.

4.3.2 ATTENDANCE MODULE

Stores one row per student, offering, date and period, with the statuses Present, Absent, On duty and Medical. Faculty may mark one period or a whole day. The module enforces the authority rules of Figure 3.4 and returns the register, the subject-wise percentage and the eligibility against the 75% line.

4.3.3 MARKS AND EXAMINATION MODULE

Faculty enter scores for assessments such as Internal Assessment 1 and 2, Assignment and Unit Test and, for laboratories, Observation, Record, Model Exam and Viva. Each assessment has its own maximum mark. Entries stay

as drafts until published, a blank field means unchanged, and students see university results, grades and CGPA computed from credits.

4.3.4 FEES, SERVICES AND LEAVE MODULE

Presents an itemised fee statement with simulated payment and a receipt number, accepts campus service requests for the administrator to resolve, and handles leave and on-duty applications with proof. Approval of an on-duty application creates the matching entries in the attendance register.

4.3.5 ADMINISTRATION AND AUDIT MODULE

Lets the administrator create departments, programmes, class sections, coordinators, subject offerings and timetable slots, manage access, and read the audit trail of privileged actions.

4.3.6 INTELLIGENCE AND CAREER HUB MODULE

The FastAPI service computes the risk and placement-readiness scores. The Career Hub offers a PDF resume analyzer that reads the file in the browser, a GitHub analyzer, skill-gap analysis, courses, certifications, project blueprints and a study assistant.

4.4 ALGORITHM / PSEUDOCODE

Algorithm 1: marking attendance for one period.

```
INPUT: faculty user F, offering O, date D, period P, list of (student,
status)
1.  if role(F) is not FACULTY or ADMIN then reject with 403
2.  teaches      <- (O.faculty == F)
3.  coordinates  <- (O.section.coordinator == F)
4.  if not (teaches or coordinates or role(F) == ADMIN) then reject with
403
5.  for each (student S, status T) in the list:
6.      if S is not in O.section then reject with 400
7.      row <- find attendance(S, O, D, P)
8.      if row exists then row.status <- T else create row(S, O, D, P, T)
9.  save all rows in one transaction
10. return the updated count
```

Algorithm 2: risk score.

```
INPUT: attendance a (%), CGPA g, IA trend t, eligibility shortfall e
1.  A <- penalty(a)      // grows steeply below the 75% line
2.  G <- penalty(g)      // lower CGPA gives a higher penalty
3.  T <- penalty(t)      // a falling internal-assessment trend scores
higher
4.  E <- penalty(e)      // papers below 75% attendance
5.  score <- 0.35*A + 0.30*G + 0.20*T + 0.15*E
6.  band  <- LOW if score < 35, MEDIUM if score < 60, else HIGH
7.  return score, band, list of the factors that contributed
```

4.5 TOOLS USED

4.5.1 DATASET

The data is a trial data set created by the application's own seeder, not imported from the college's records: five departments, ten classes, 59 students, 25 staff members and 28,497 attendance records, together with marks, university results, fee items, leave applications, companies and internships. The seeder writes the credentials of the generated accounts to a file that is excluded from version control. The screenshots in this report show this trial data.

4.5.2 IMPLEMENTATION DETAILS

The server runs on port 8081, the front end on 5173 and the scoring service on 8000. Three controllers expose the functionality under /api/student, /api/faculty and /api/admin, and the authentication controller handles /api/auth. Table 4.4 lists selected routes.

TABLE 4.4 SERVER ENDPOINTS

Route	Method	Purpose
/api/auth/login	POST	Password sign-in; returns token and mustChangePassword
/api/auth/login-code	POST	Quick-code sign-in for students
/api/auth/change-password	POST	Change password (minimum 8 characters)
/api/faculty/attendance/period	POST	Mark one period
/api/faculty/attendance/day	POST	Mark a whole day
/api/faculty/marks	GET, POST	Read or save an assessment, draft or published
/api/faculty/leave-applications/{id}/approve	POST	Approve a leave or OD application
/api/student/attendance	GET	Register, per-subject and per-day roll-ups
/api/student/fees and /fees/pay	GET, POST	Statement and simulated payment
/api/admin/users	GET, POST	List or provision accounts
/api/admin/audit	GET	Most recent privileged actions

4.6 SNAPSHOTS

The following snapshots were taken from the running application at 1440 x 900 using the seeded trial data. The student screens belong to the account of the first author (CSE II-E); the faculty screens belong to the class coordinator of the same class.

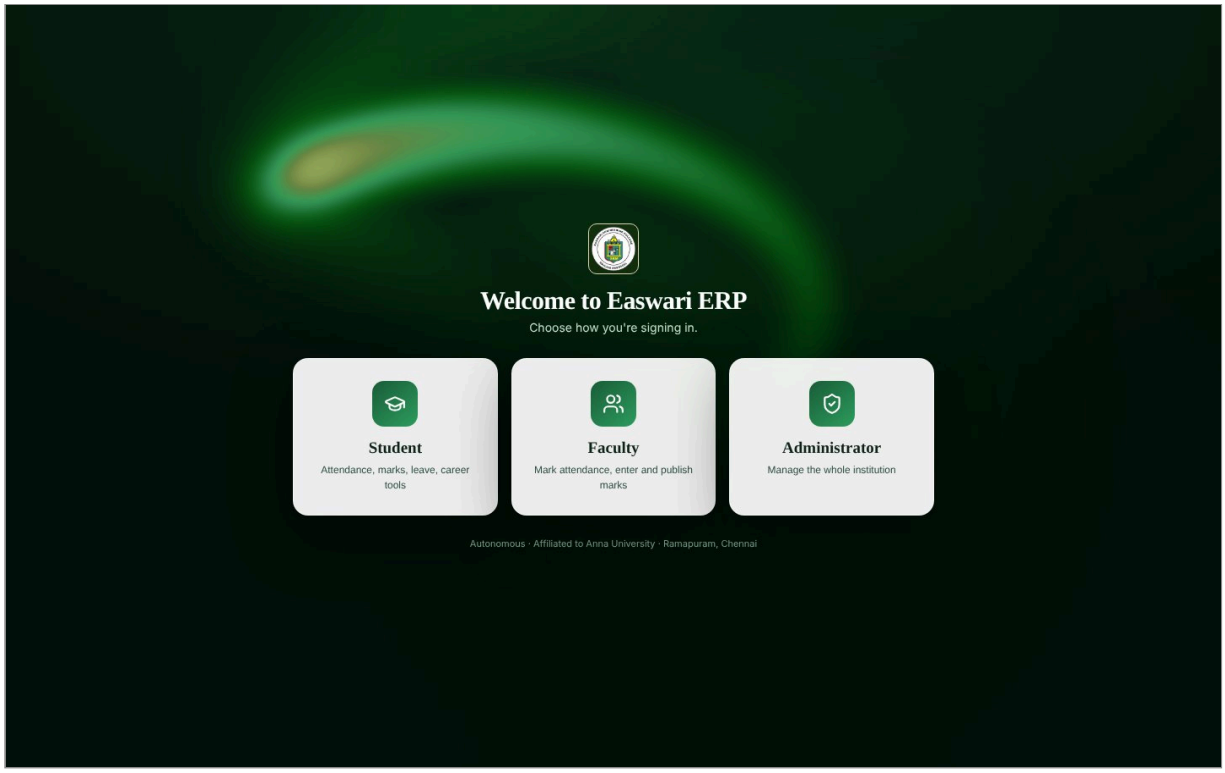


FIGURE 4.7 WELCOME AND ROLE SELECTION

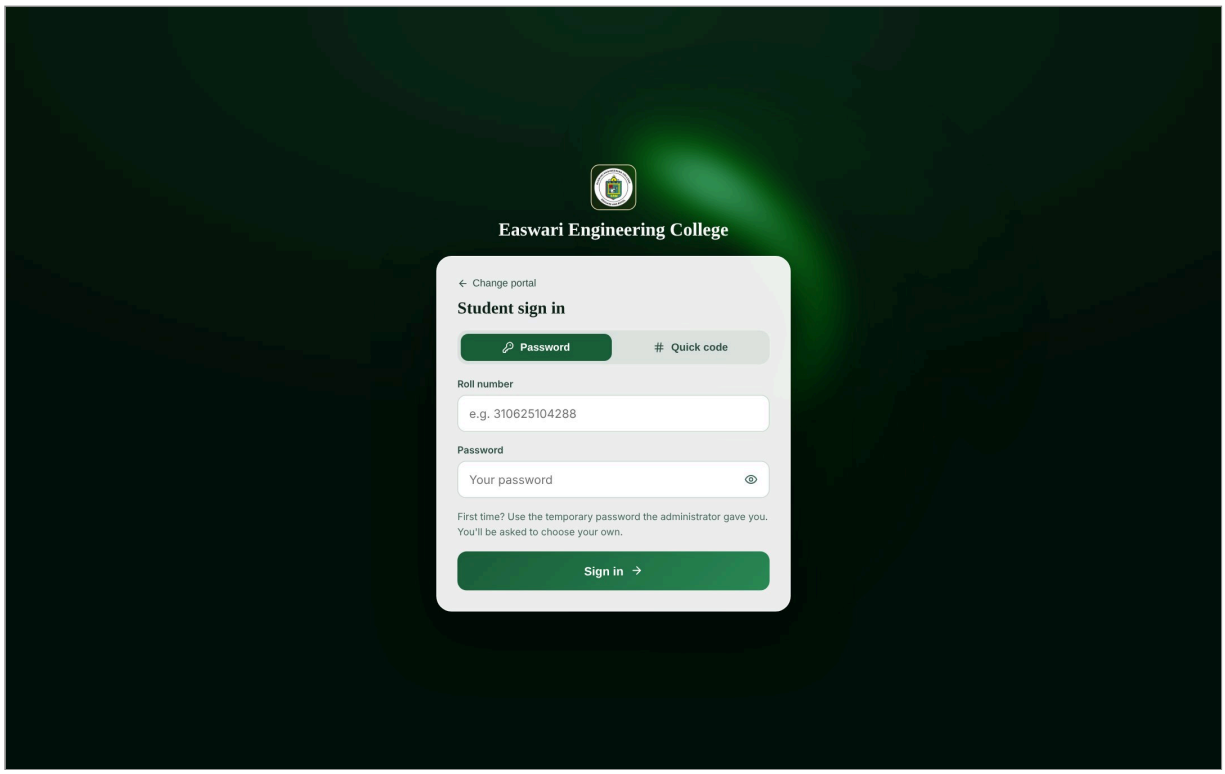


FIGURE 4.8 STUDENT SIGN-IN

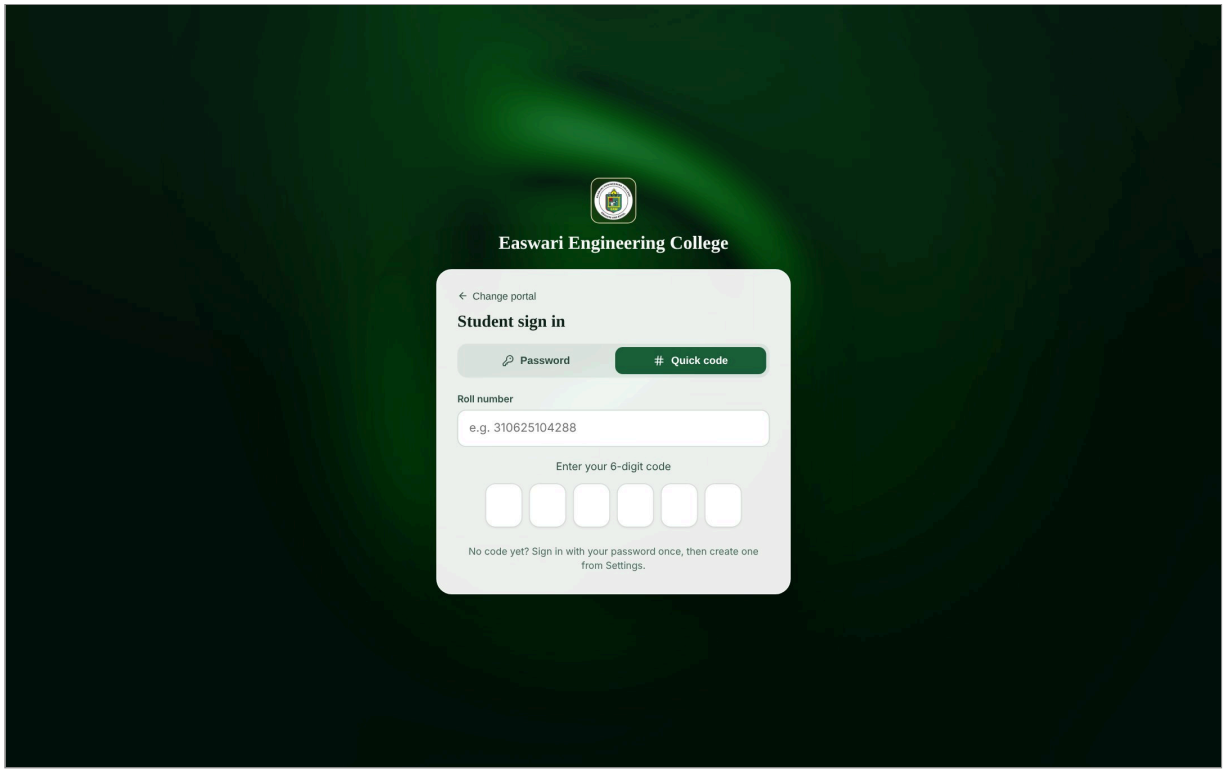


FIGURE 4.9 QUICK-CODE SIGN-IN

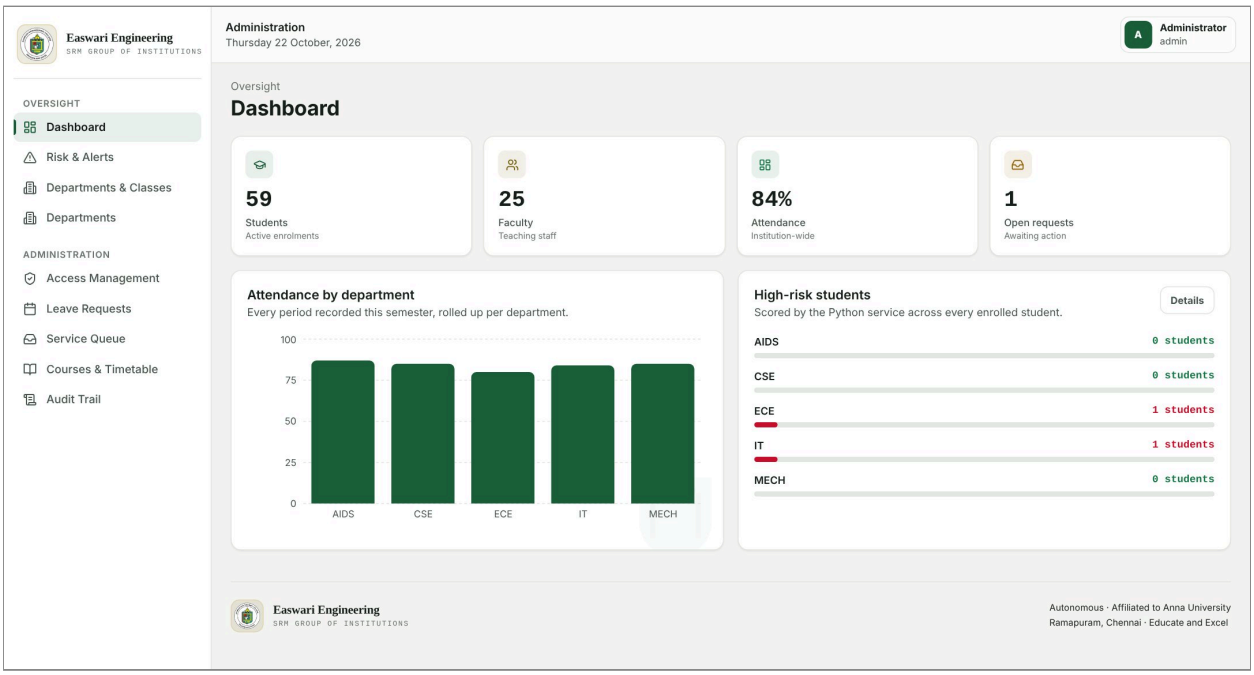


FIGURE 4.10 ADMINISTRATOR DASHBOARD

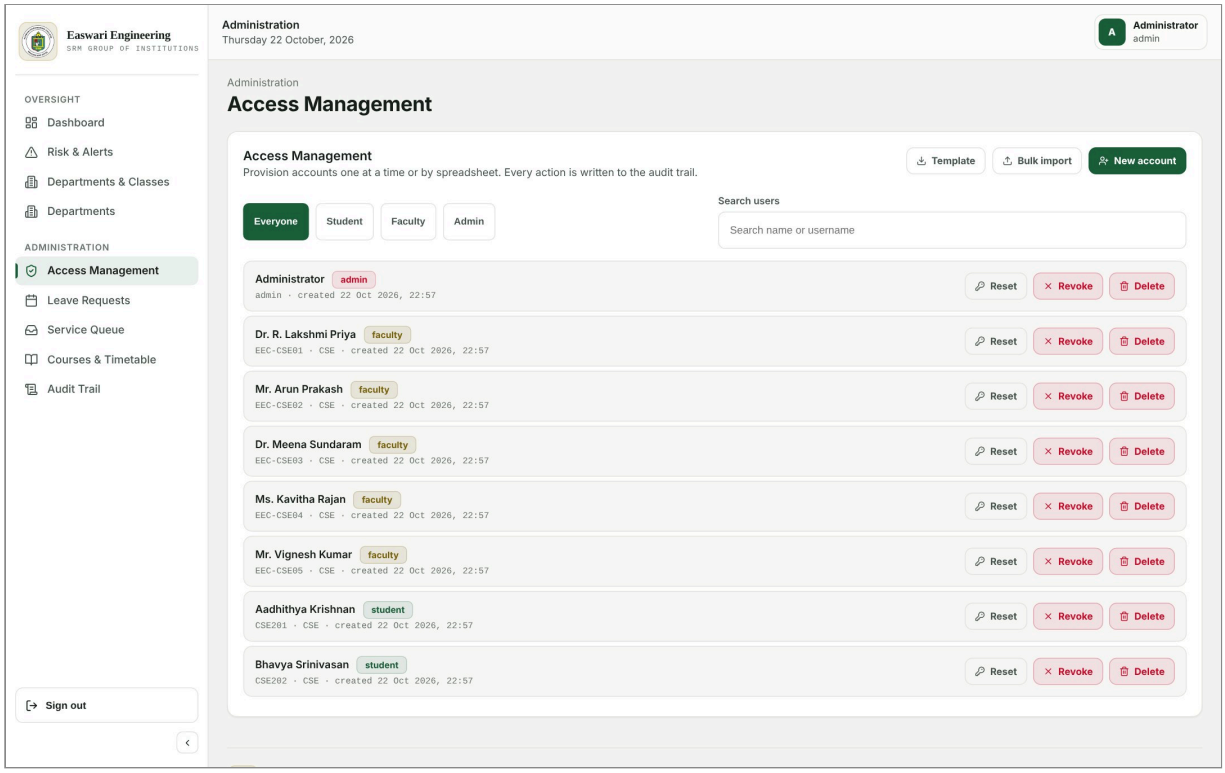


FIGURE 4.11 ACCESS MANAGEMENT

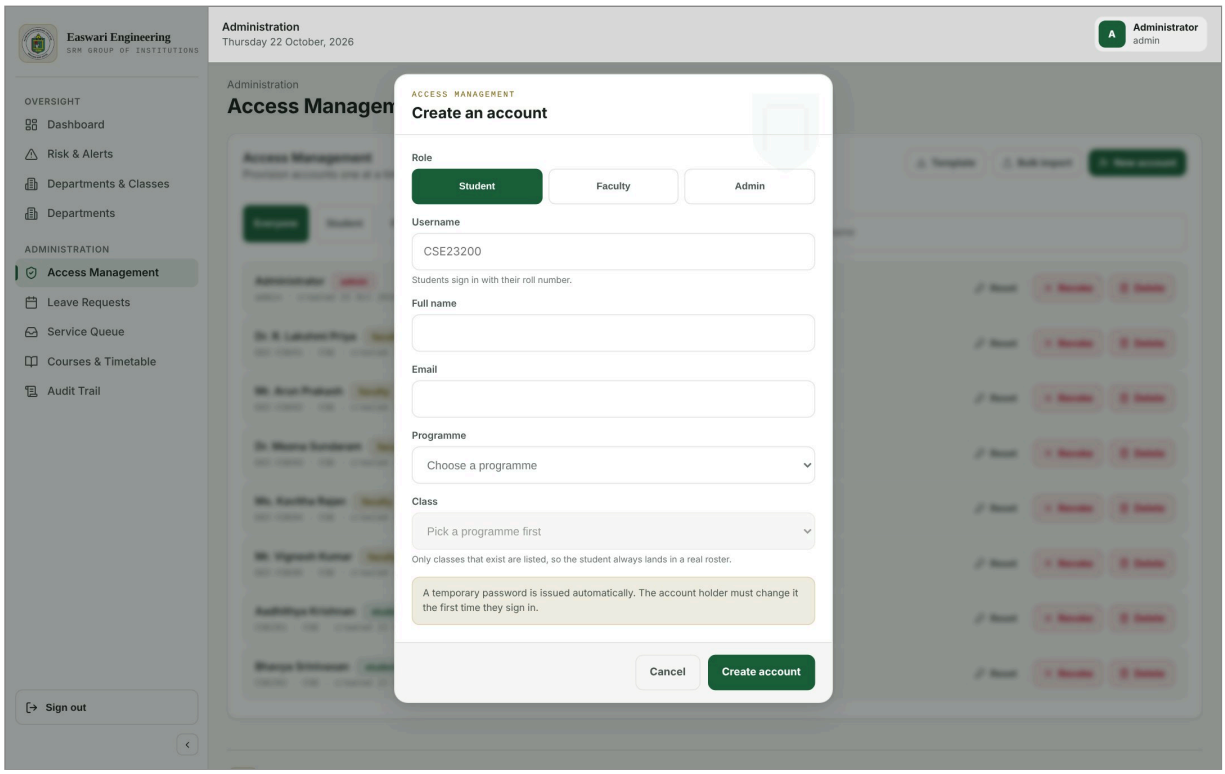


FIGURE 4.12 CREATE AN ACCOUNT

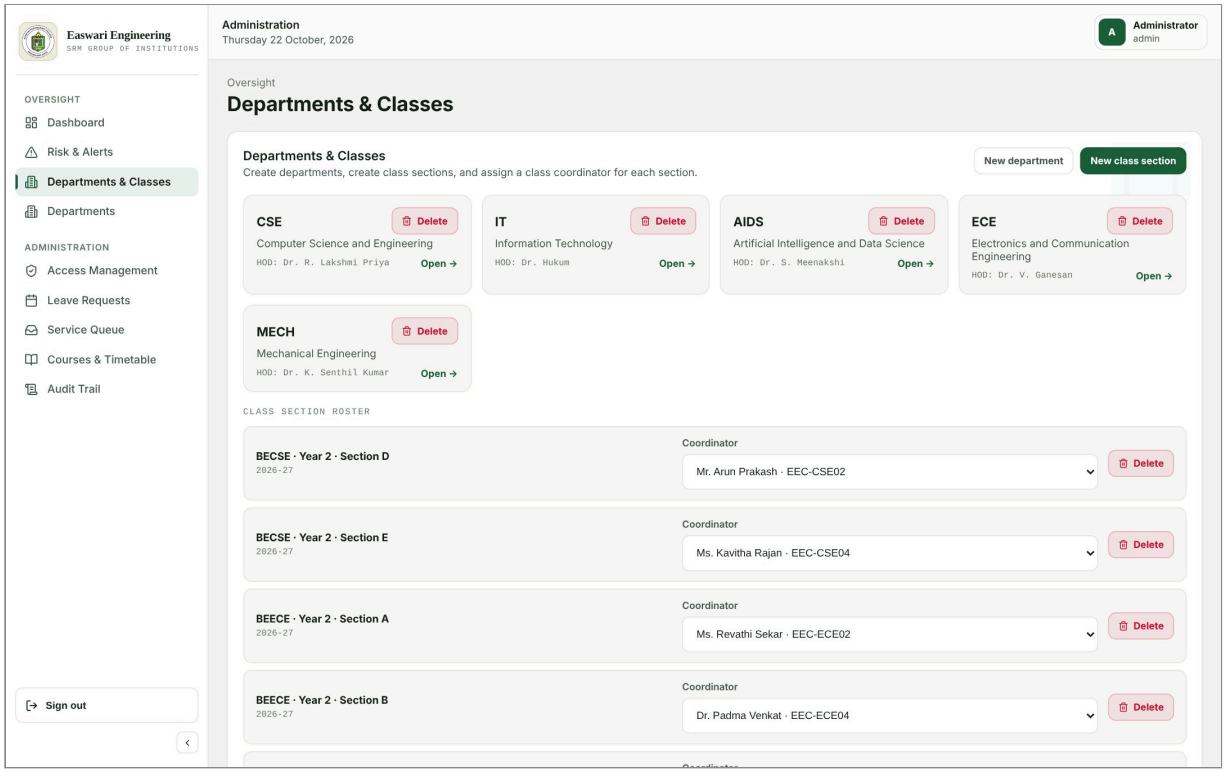


FIGURE 4.13 DEPARTMENTS AND CLASSES

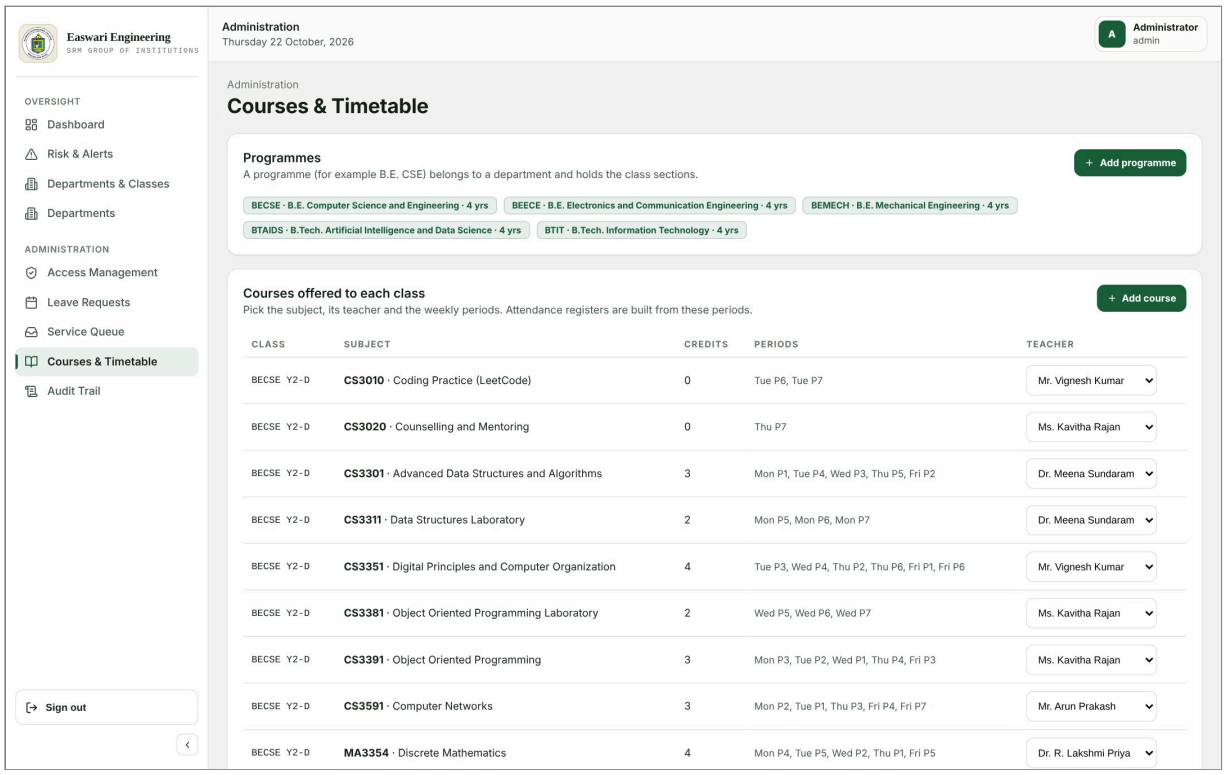


FIGURE 4.14 COURSES AND TIMETABLE

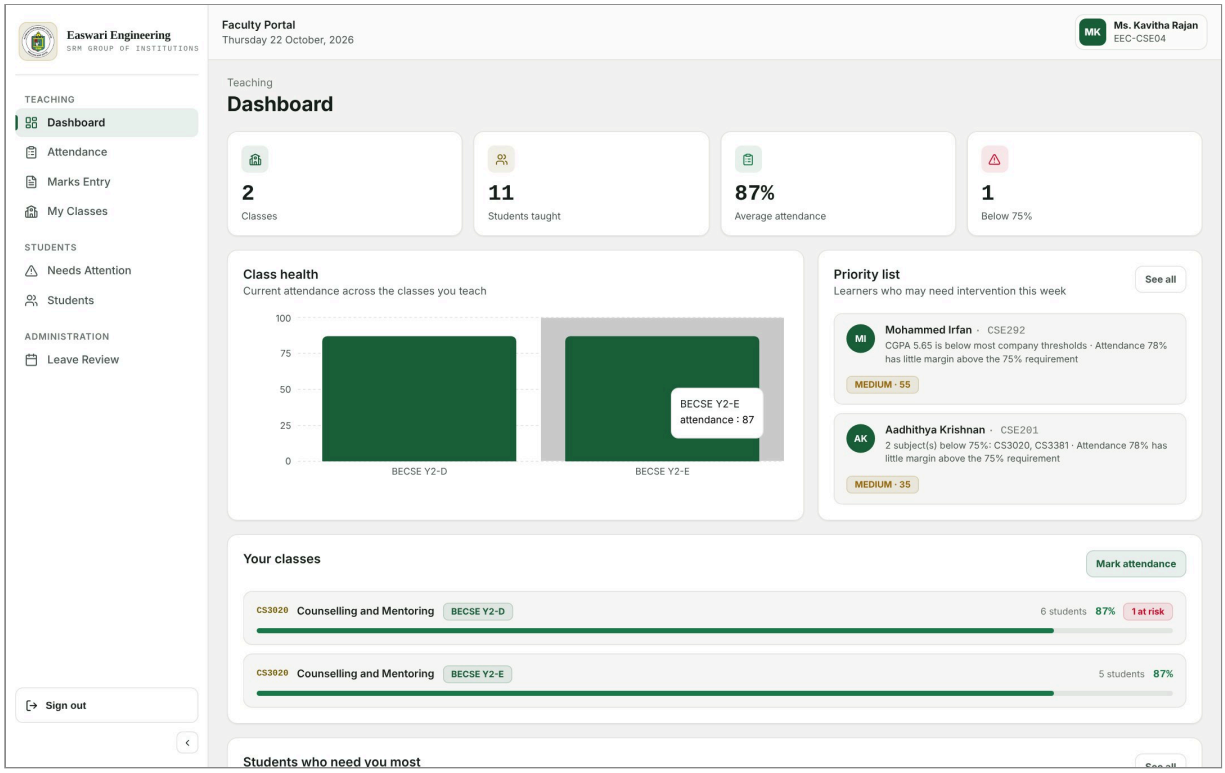


FIGURE 4.15 FACULTY DASHBOARD

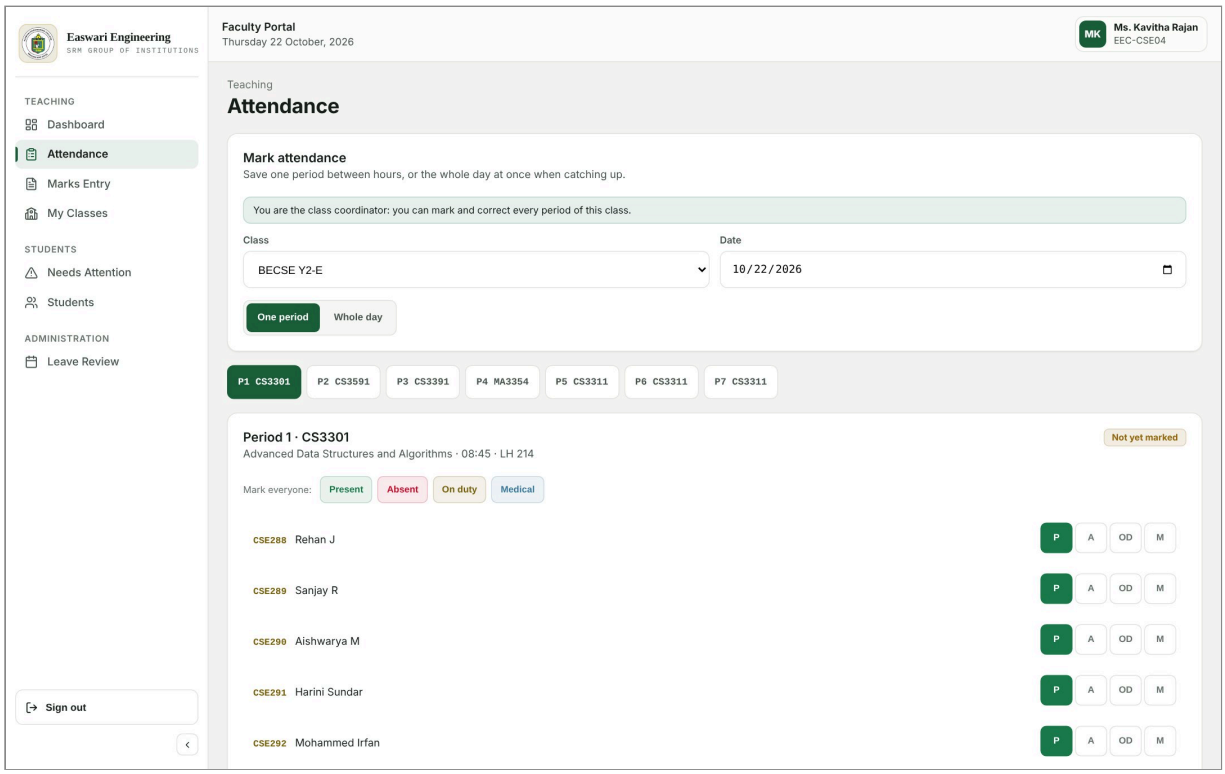


FIGURE 4.16 ATTENDANCE ENTRY BY THE CLASS COORDINATOR

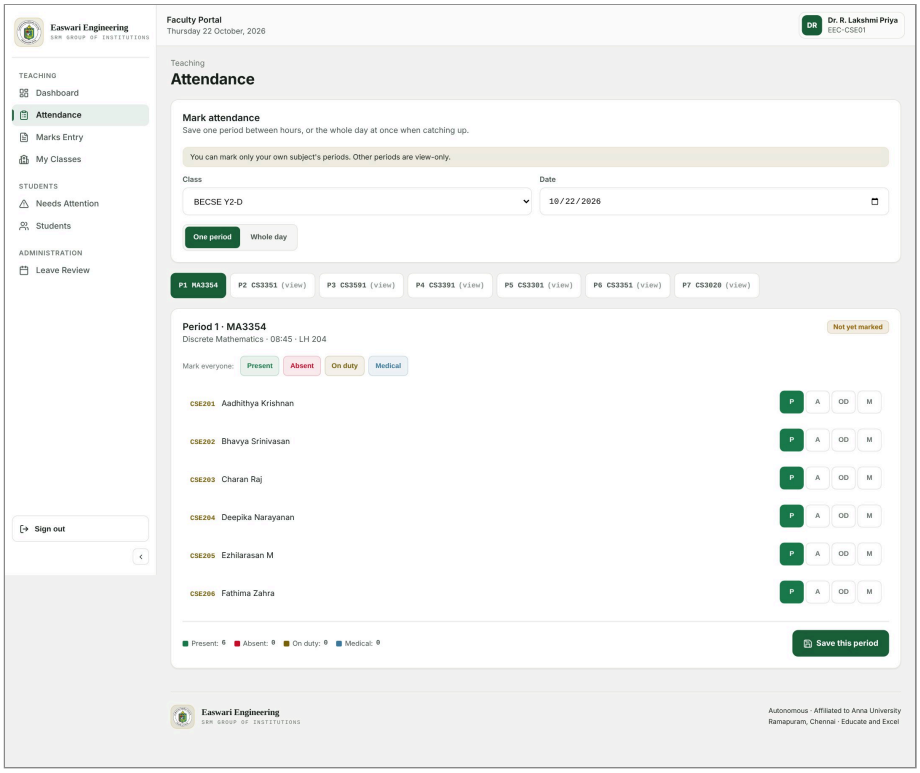


FIGURE 4.17 ATTENDANCE ENTRY BY A SUBJECT TEACHER

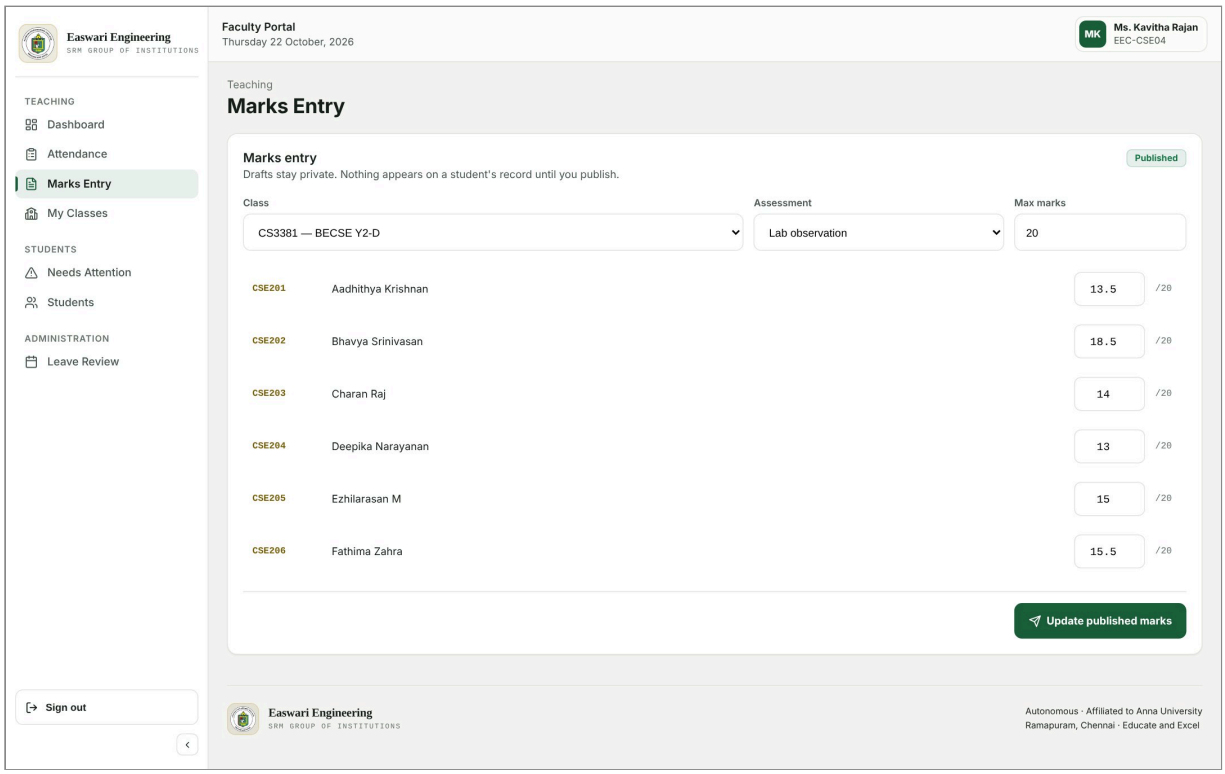


FIGURE 4.18 MARKS ENTRY

4.7 SUMMARY

This chapter described the methodology of the Easwari ERP Portal: the hardware and software, the division of work, the six modules, the algorithms, the tools and the dataset, and snapshots of the running application. The next chapter presents the testing and the results.

CHAPTER 5
TESTING AND RESULTS

5.1 TESTING STRATEGIES

Testing validated both the correctness of each rule and the behaviour of the running system. A multi-level strategy was adopted:

1. **UNIT TESTING:** eleven JUnit tests cover the attendance, marks, faculty and password rules, including the generator of temporary passwords.
2. **INTEGRATION TESTING:** the endpoints of the packaged server were called with the administrator, the class coordinator, a subject teacher and a student account, and the responses were recorded.
3. **FUNCTIONAL TESTING:** a real browser was driven through the sign-in flow and every portal screen to confirm that the information reaches the person who needs it.
4. **SECURITY TESTING:** wrong passwords, missing or forged tokens, access to another student’s record and teachers acting outside their subject were attempted and had to be refused.
5. **ROBUSTNESS TESTING:** blank fields, duplicate usernames, values above the maximum mark and unknown status values were submitted to check the error messages.

5.2 TEST CASES AND EXECUTION

The test cases were executed on 22 October 2026 against the packaged application running with the fully seeded institution. Table 5.2.1 lists all 68 test cases with the input, the expected output, the actual output recorded during the run and the status. 67 passed and 1 revealed a defect.

TABLE 5.2.1 TEST CASES AND RESULTS

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC1	Administrator signs in with the correct password	200, token issued, role ADMIN	200, token issued, role ADMIN	Pass
TC2	Administrator signs in with a wrong password	401 with a neutral message	401 – “That username and password combination isn’t recognised”	Pass
TC3	Unknown username	401, same neutral message	401, same message (the response does not reveal which part was wrong)	Pass
TC4	Blank username and password	400 naming both fields	400 – “username: must not be blank; password: must not be blank”	Pass
TC5	Administrator creates a student account	200, random temporary password issued	200, user created, 10-character temporary password returned once	Pass
TC6	Create a second account with the same username	409 conflict	409 – “That username is already in use”	Pass
TC7	Create an account without a name	400 naming the field	400 – “name: must not be blank”	Pass
TC8	New student signs in with the temporary password	200 with mustChangePassword true	200, mustChangePassword = true	Pass
TC9	Change password to a 5-character value	400	400 – “Use at least 8 characters”	Pass

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC10	Change password with a wrong current password	400	400 – “Current password is incorrect”	Pass
TC11	Change password correctly, then sign in again	200, flag cleared	200 – “Password updated”; next sign-in shows mustChangePassword = false	Pass
TC12	Create a quick code with 5 digits	400	400 – “The code must be exactly 6 digits”	Pass
TC13	Create a 6-digit quick code, then sign in with the code	200 both times	200 – “Quick code saved”; code sign-in returned a token	Pass
TC14	A faculty account tries to create a quick code	Refused	400 – “The quick code is available for student accounts only”	Pass
TC15	Six consecutive wrong codes for one student	Locked after the fifth	Responses 401, 401, 401, 401, 401, then 429	Pass
TC16	Correct code while the lock is active	429, advice to use the password	429 – “Too many wrong codes. Sign in with your password instead, or try again in a few minutes.”	Pass
TC17	Class coordinator marks a period of a subject taught by another teacher	200 (coordinator controls the whole class)	200 – “Attendance saved for period 3”	Pass

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC18	Subject teacher marks a period of his or her own subject in the same class	200	200 – “Attendance saved for period 2”	Pass
TC19	Same subject teacher marks a period of another teacher’s subject	403	403 – “You are not assigned to this class”	Pass
TC20	Subject teacher’s class list shows editable flags	Only own subject editable	Eight subjects returned editable = false and the teacher’s own subject editable = true	Pass
TC21	Mark the same period again for the same students	Update, no duplicate	200; the unique slot constraint turned it into an update and the student’s attendance stayed at 91%	Pass
TC22	Period number 9	400	400 – “period: must be less than or equal to 8”	Pass
TC23	Include a student from another class	400 naming the student	400 – “CSE201 is not in BECSE Y2-E”	Pass
TC24	Use an unknown status value	400	400 – request body could not be read	Pass
TC25	Score 25 for an assessment out of 20	400	400 – “Score 25.0 is outside 0–20”	Pass
TC26	Lower the maximum to 10 when 18 is already stored	400	400 – “A score of 18.0 is already recorded, above the new maximum of 10.”	Pass

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC27	Save a blank score for a student	Stored value unchanged	200; the sheet still showed 18.0 for that student	Pass
TC28	Subject teacher enters marks for another teacher's subject	403	403 – "You are not assigned to this class"	Pass
TC29	Coordinator enters marks for another teacher's subject	200	200 – saved	Pass
TC30	Teacher publishes a unit test for the class	200, visible to students	200 – "Marks published and visible to students"	Pass
TC31	Student opens the assessments list afterwards	200 with the components	200; components and maximum marks listed for every subject	Pass
TC32	Edit an already published assessment	Saved, student told	200 – "Saved. This assessment is already published, so students see the change."	Pass
TC33	Student opens the fee statement	200 with totals	200 – billed 2,28,200; paid 2,04,000; due 24,200; overdue 16,000; 11 items	Pass
TC34	Student pays one item (UPI)	200, receipt issued, totals change	200; paid rose by 750 and due fell by 750	Pass
TC35	Pay the same item again	409	409 – "CSE association and symposium fee is already paid."	Pass
TC36	Pay with an empty list of items	400	400 – "feeIds: must not be empty"	Pass

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC37	Pay an item that belongs to another student	404	404 – “That fee isn’t on your statement.”	Pass
TC38	Student submits an on-duty application	200, status PENDING	200, status PENDING	Pass
TC39	From-date later than to-date	400	400 – “From date must be on or before to date”	Pass
TC40	Coordinator opens the pending list	Contains the new application	200; application listed with two older pending ones	Pass
TC41	Staff of the class approve it	200, APPROVED	200 – approved	Pass
TC42	Approve the same application again	409	409 – “This application is no longer pending”	Pass
TC43	Student raises a library request	200, ticket number, OPEN	200 – ticket EEC-15043, status OPEN	Pass
TC44	Administrator resolves it with a note	200, RESOLVED	200 – RESOLVED	Pass
TC45	Administrator sends an unknown status value	400 with a clear message	500 – generic server message	Fail (defect D1)
TC46	Apply to a drive open to ECE only	403 with the reason	403 – “You don’t meet the criteria: Open to ECE only”	Pass
TC47	Apply to an eligible drive	200	200 – “Application submitted”	Pass

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC48	Apply to the same drive again	409	409 – “You have already applied to this company”	Pass
TC49	Student calls an administrator route	403	403	Pass
TC50	Student calls a faculty route	403	403	Pass
TC51	Faculty calls a student route	403	403	Pass
TC52	Request without a token	Refused	403	Pass
TC53	Request with a forged token	Refused	403	Pass
TC54	Teacher of another department opens a class roster	403	403 – “You are not assigned to this class”	Pass
TC55	Invalid JSON body at sign-in	400	400 – “The request body couldn’t be read. Check the fields and try again.”	Pass
TC56	Unknown route	404	404 – “That page or endpoint doesn’t exist.”	Pass
TC57	Wrong HTTP method	405	405 – “That action isn’t supported here.”	Pass
TC58	Administrator revokes a student’s access	200	200	Pass
TC59	Revoked student signs in	Refused with a clear message	403 – “This account is deactivated. Contact the administration office.”	Pass

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC60	Administrator grants access again; student signs in	200	200 for both	Pass
TC61	Administrator resets the password	Temporary password, change forced	200 – “Share this once. The user must change it at next sign-in.” (the temporary value is the role default; see D2)	Pass
TC62	Administrator deletes the test account	200	200	Pass
TC63	Reassign the class coordinator of CSE II-E	200	200, class summary returned	Pass
TC64	Edit a department’s name	200	200, department returned	Pass
TC65	Read the audit trail	Actions listed newest first	200 – 23 entries including DEPARTMENT_UPDATED, CLASS_COORDINATOR_UPDATED, USER_DELETED, PASSWORD_RESET, ACCESS_GRANTED, ACCESS_REVOKED	Pass
TC66	Scoring service stopped; open the administrator dashboard and Risk page	Pages render without an error	Dashboard and Risk page rendered; risk counts showed zero; health reported scoringService DOWN	Pass
TC67	Scoring service started; open the Risk page and the learning path	Real results	Risk page listed one high-risk student in ECE and one in IT; learning path listed three recommendations with reasons	Pass

Test Case ID	Input Type	Expected Output	Actual Output	Status
TC68	Copy the database file, stop and restart the server on the copy	All records identical	Users 85 = 85; attendance of the student 499 periods, 454 attended, 91% (unchanged); fees billed, paid and due unchanged; the seeder did not run again	Pass

The failing case (TC45) showed that an unknown status value sent to the service-request update route returns a generic 500 response instead of a 400 message. A second observation from the security tests was that the administrator’s password reset issues a fixed role default instead of a random value. Both are recorded as limitations in Chapter 6 together with the recommended fixes.

5.3 RESULTS AND SCREENSHOTS

The system was started from a single packaged server file in about nine seconds and created the full trial institution in about ten seconds. Table 5.3.1 summarises the quantitative results.

TABLE 5.3.1 QUANTITATIVE RESULTS

Metric	Value
REST test cases executed	68 (67 passed, 1 defect)
Unit tests	11 of 11 passed
Server start-up time	About 9 seconds
Time to seed the full institution	About 10 seconds
Seeded records	5 departments, 10 classes, 59 students, 25 staff, 28,497 attendance records
Screens documented	40 screens across the three portals
Executable jar size	About 72 MB

The central claim of the design, that the authority to change attendance and marks is enforced by the server, was confirmed: a coordinator can mark and enter marks for every subject of the class, a subject teacher is refused for any other subject with a 403 response, and a teacher who has no connection with the class cannot even read its roster. Marking a period twice updated the existing row, so the student’s percentage did not change. The leave workflow, the fee payment and the placement eligibility rules each refused the invalid case with a clear message.

TABLE 5.3.2 RISK SCORE FOR THREE PROFILES

Profile	Attendance	CGPA	Score	Band
Healthy	93%	8.90	0	LOW
Borderline	78%	7.20	36	MEDIUM
Struggling	71%	6.40	66	HIGH

The risk score rises steeply when attendance falls below the 75% line, as designed. The student, faculty and administrator screens that show these results are given below.

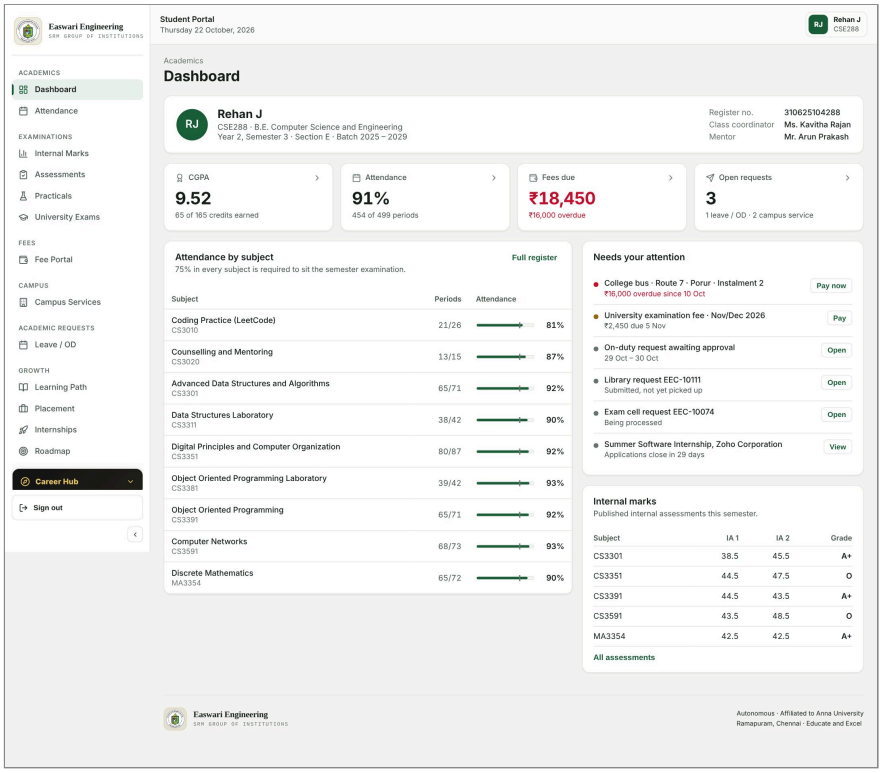


FIGURE 5.1 STUDENT DASHBOARD

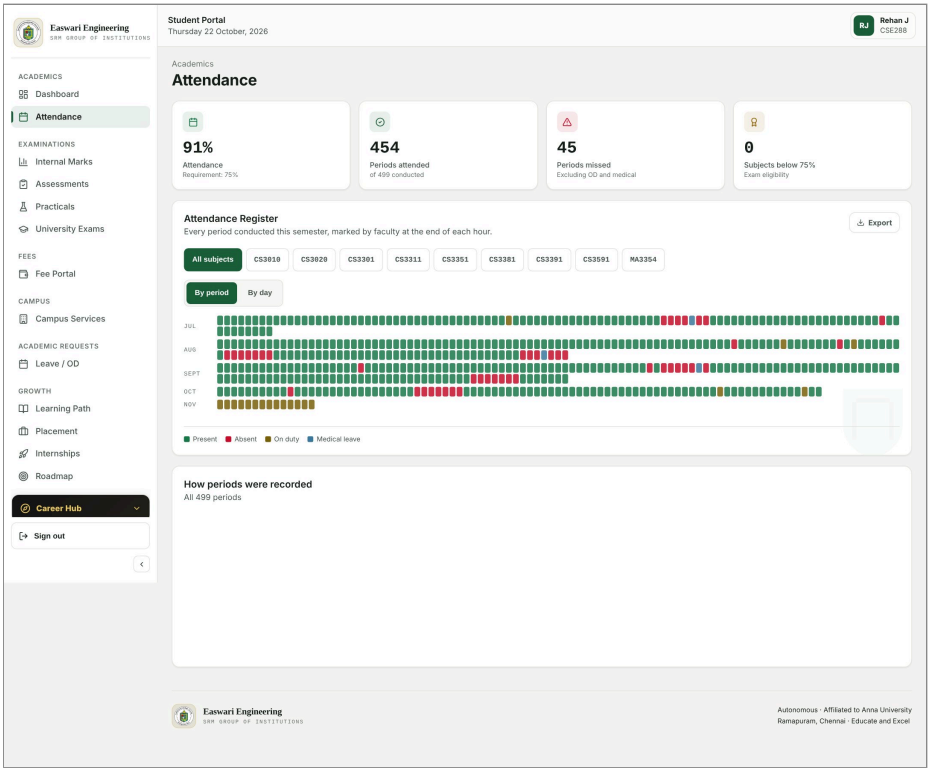


FIGURE 5.2 ATTENDANCE REGISTER

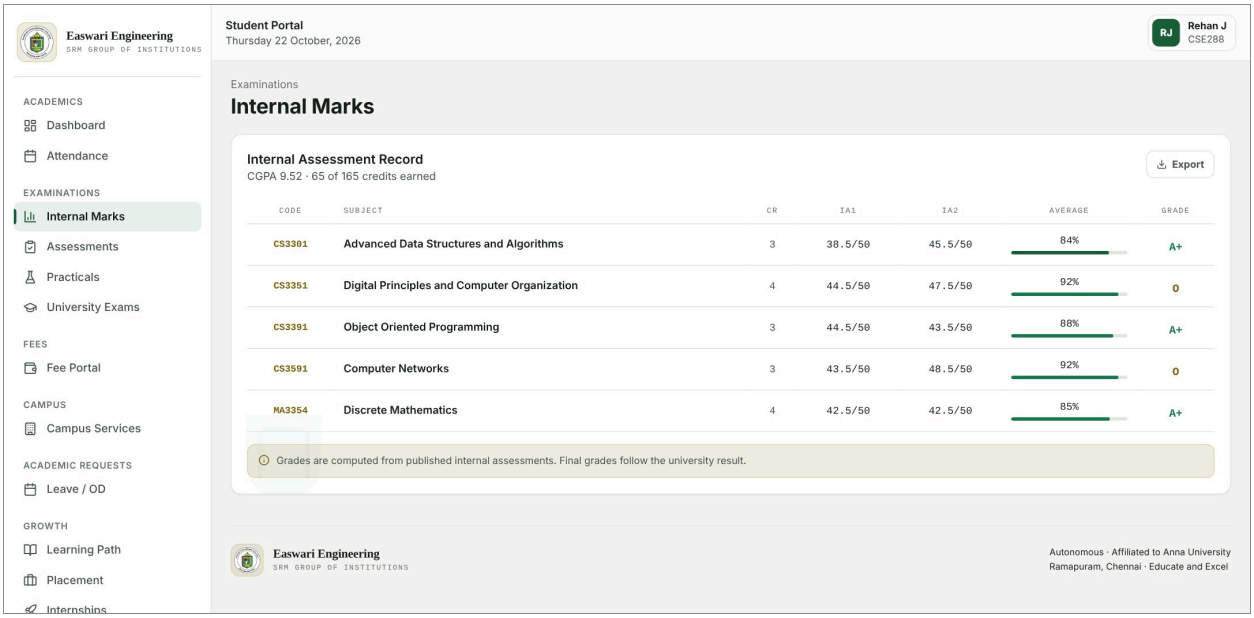


FIGURE 5.3 INTERNAL MARKS

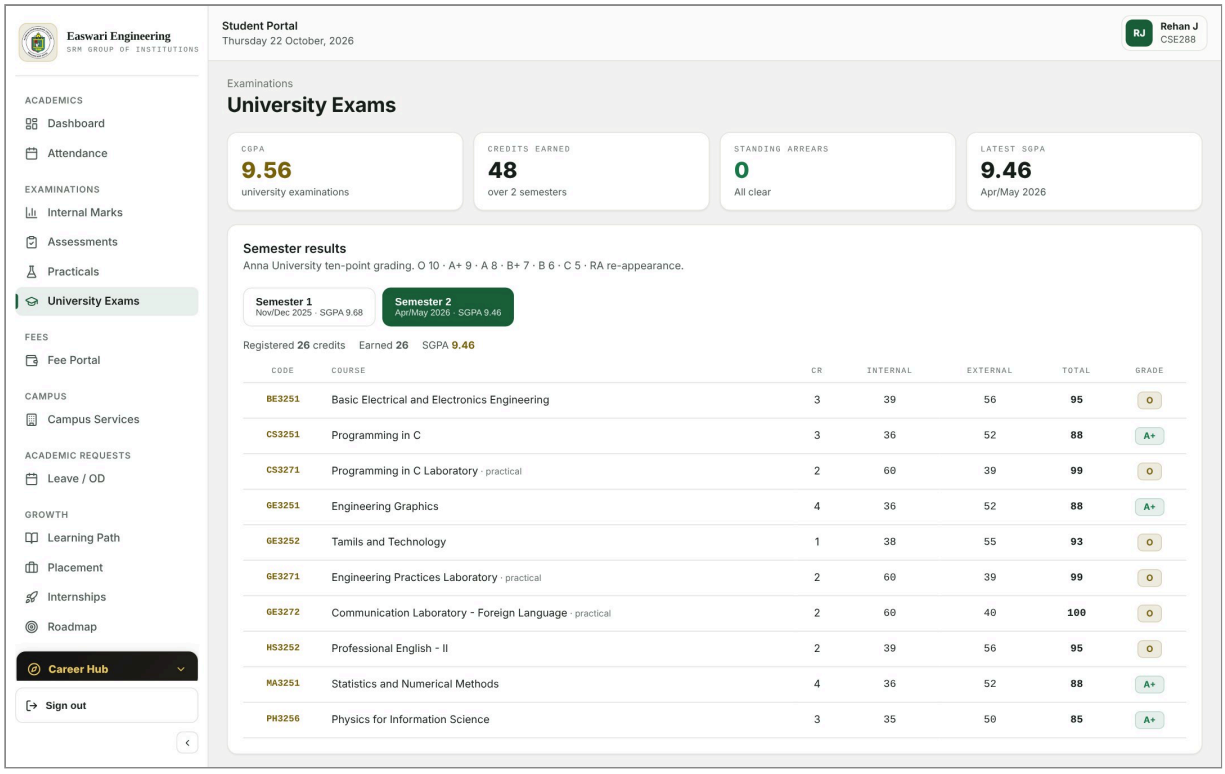


FIGURE 5.4 UNIVERSITY EXAMINATION RESULTS

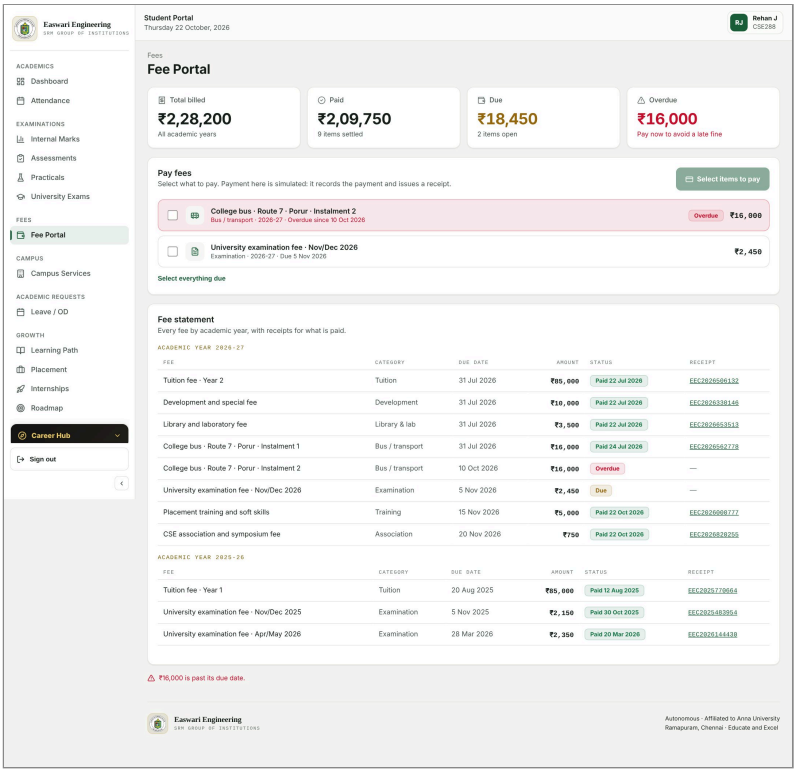


FIGURE 5.5 FEE PORTAL

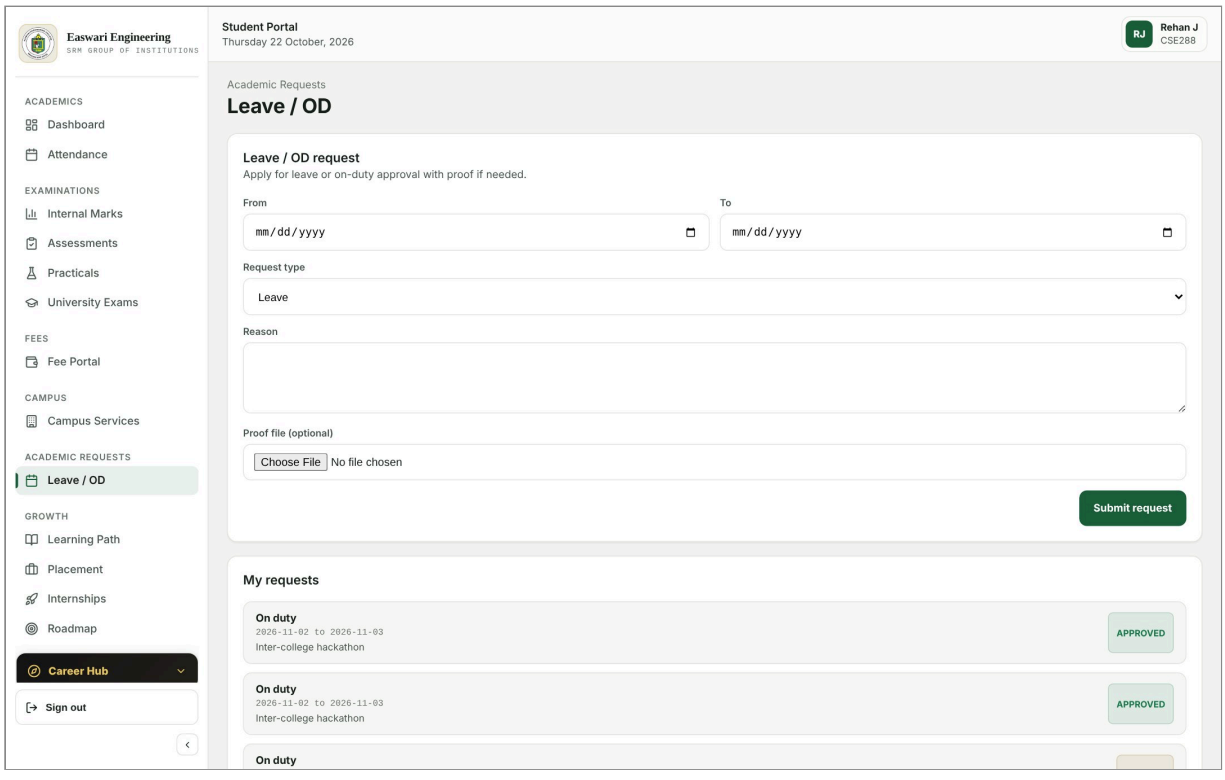


FIGURE 5.6 LEAVE AND ON-DUTY APPLICATION

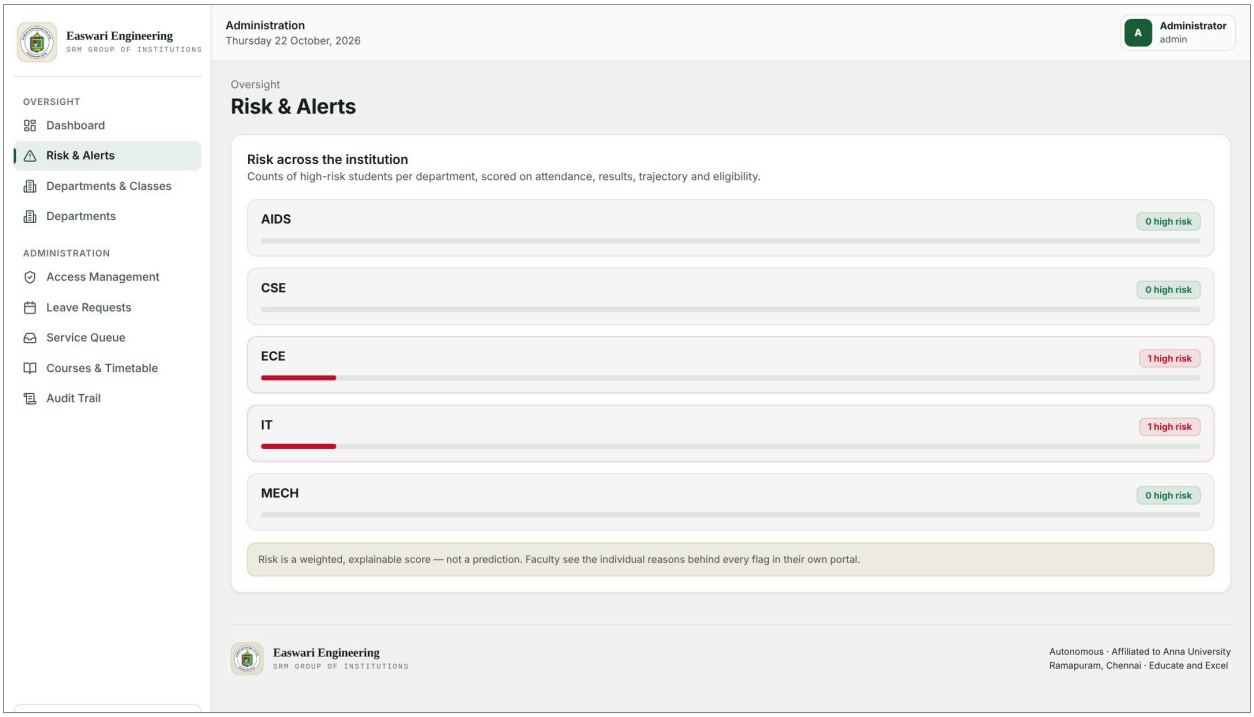


FIGURE 5.7 RISK AND ALERTS

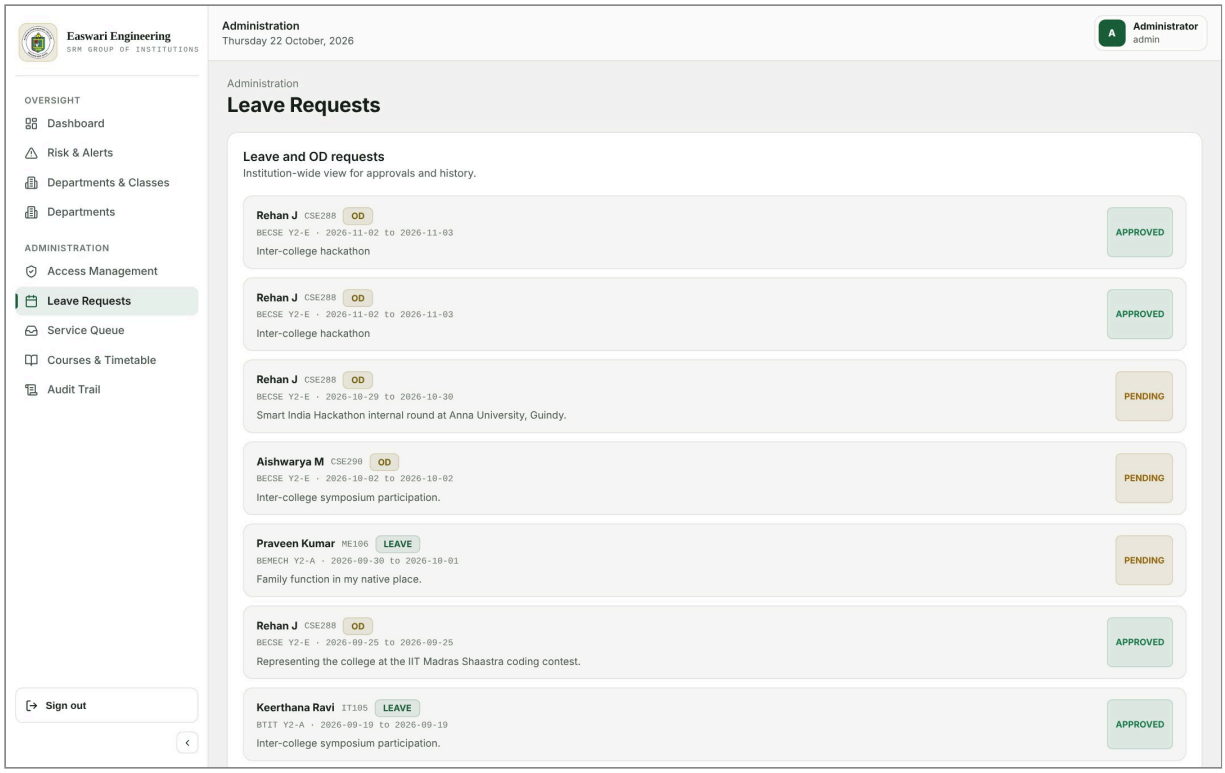


FIGURE 5.8 LEAVE REQUESTS (ADMINISTRATION)

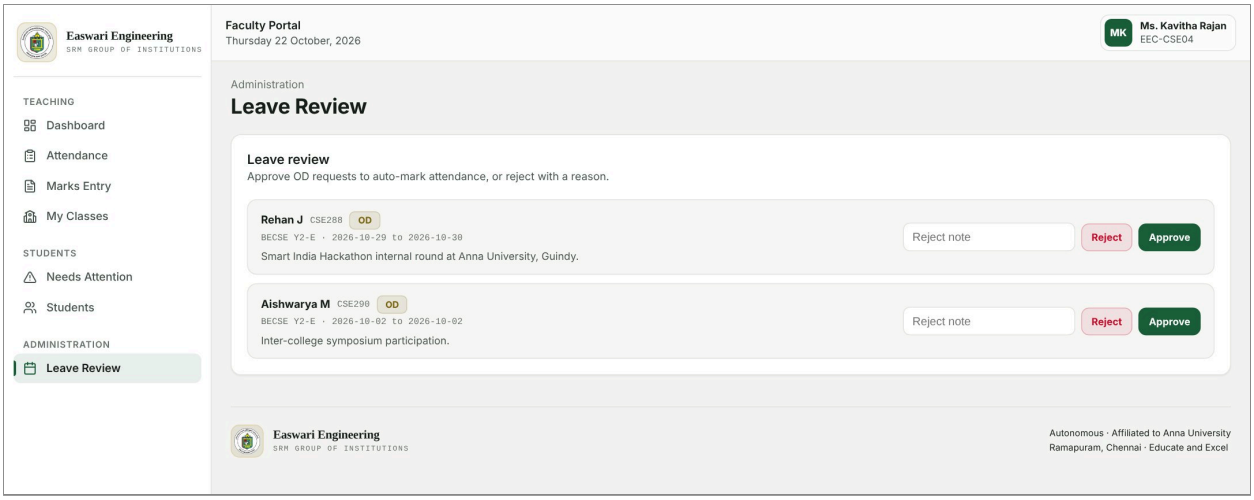


FIGURE 5.9 LEAVE REVIEW

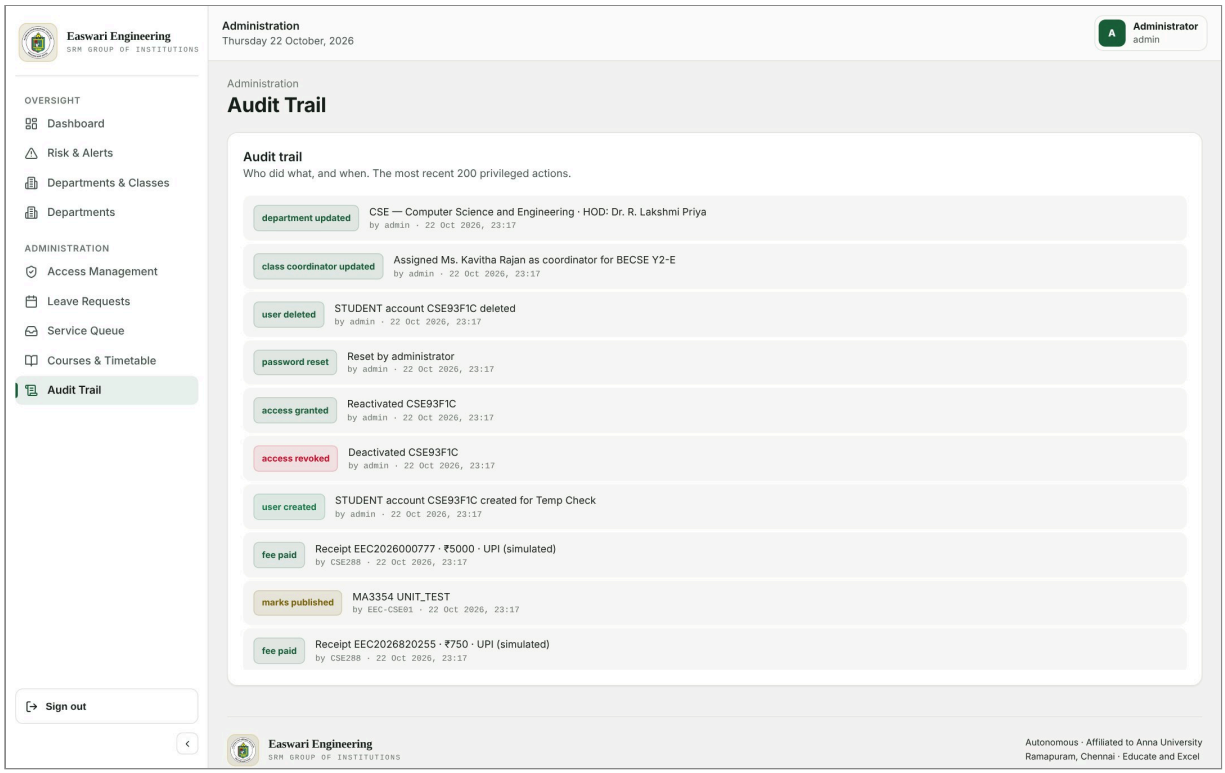


FIGURE 5.10 AUDIT TRAIL

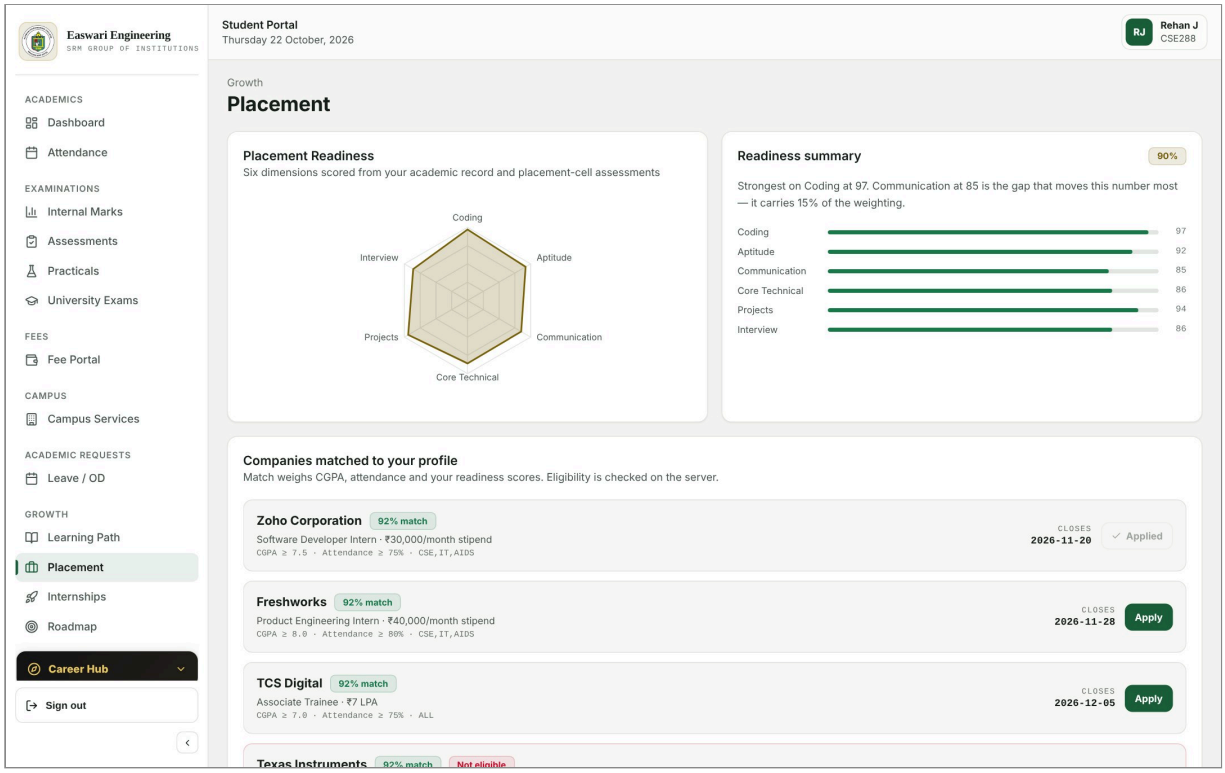


FIGURE 5.11 PLACEMENT

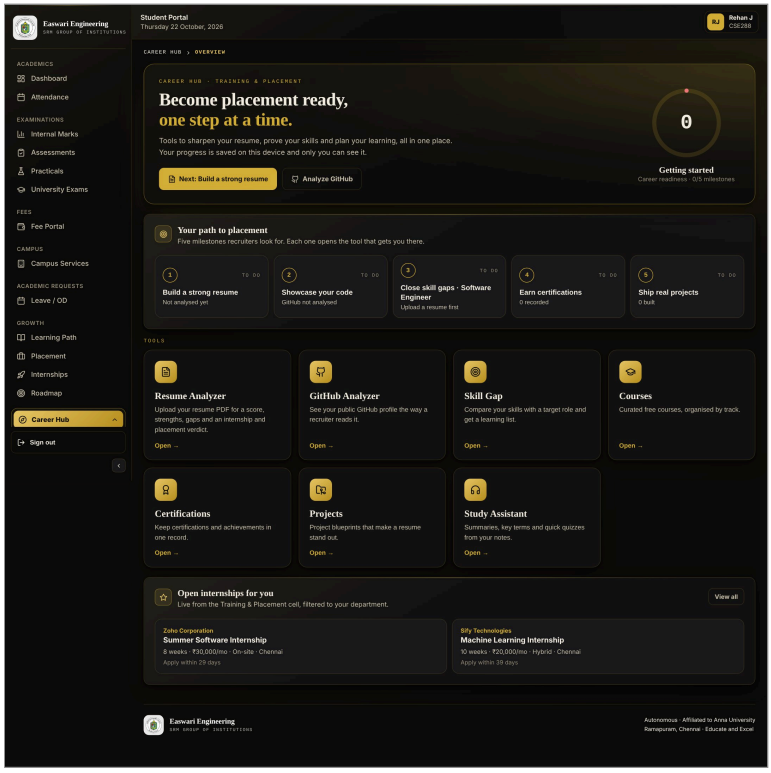


FIGURE 5.12 CAREER HUB OVERVIEW

CHAPTER 6
CONCLUSION

6.1 INTRODUCTION

This project set out to give a college a single, controlled record of attendance and internal assessment, and to turn that record into timely information for students, faculty and administrators. Existing practice relied on paper registers and spreadsheets that drift apart and that cannot say who changed what.

The Easwari ERP Portal addresses these problems with period-level attendance under a database constraint, server-enforced authority rules, a publish gate for marks, self-service views for students and an explainable risk score. This chapter summarises what was achieved, the limitations and the scope for future work.

6.2 CONCLUSION

The platform shows that attendance control and student progress monitoring can be handled in one role-based system in which each fact is stored once and each rule is enforced where it cannot be bypassed. Students have one place to see their attendance, marks, results, fees and requests; faculty have a quick way to mark and publish; the class coordinator has authority over the whole class; and the administrator has oversight and an audit trail.

In 68 recorded test cases 67 behaved as expected, and all eleven unit tests passed. The limitations are stated plainly:

- Payment is simulated and no payment gateway is connected.
- Defect D1: an unknown status value sent to the service-request update route returns a generic 500 response instead of a 400 message.
- Defect D2: the administrator’s password reset issues a role default value; it should use the random generator used for new accounts.

- The signing secret has a development default and the H2 console route is permitted; both must be changed before production use.
- The data shown is a seeded trial data set, and the risk score is a transparent weighted heuristic whose weights need calibration with real outcomes.

6.3 SUMMARY AND FUTURE SCOPE

This project covered the full path of a campus platform, from problem identification and literature review through design, implementation, testing and evaluation.

Key Achievements:

- Three role-based portals behind one sign-in with temporary passwords, forced first-login change and a quick-code lockout.
- Period-level attendance with a unique slot constraint and coordinator, teacher and administrator authority enforced by the server.
- Marks with per-assessment maximums and a publish gate, fees, leave, services, placement and a Career Hub.
- An explainable risk score from a separate scoring service, an audit trail, and 68 recorded test cases.

FUTURE SCOPE:

1. **PAYMENT GATEWAY:** replace the simulated payment with a gateway in test mode and then in production.
2. **REAL DATA:** import real timetables, student lists and university results instead of the seeder.
3. **ALERTS:** notify students and, on request, parents when attendance in a subject falls near 75%.
4. **HARDENING:** rate limiting for password sign-in, optional two-factor authentication, and fixes for D1 and D2.

5. **CALIBRATION:** calibrate the risk weights against real outcomes of two or three semesters.
6. **ATTENDANCE CAPTURE:** QR-code or mobile capture reconciled with the faculty entry.
7. **MOBILE AND MORE PORTALS:** an installable mobile layout and parent, mentor, hostel and library modules.

The most useful next step is to pilot the system with one real class for a semester and record the time saved and the number of students who corrected short attendance after an early alert.

SAMPLE SOURCE CODE

Attendance slot constraint – AttendanceRecord.java (Back End)

```
@Entity @Table(name = "attendance_records",
    uniqueConstraints = @UniqueConstraint(
        name = "uk_attendance_slot",
        columnNames = {"student_id", "subject_offering_id",
                       "class_date", "period"}))
public class AttendanceRecord { ... }
```

Route guards – SecurityConfig.java (Back End)

```
.requestMatchers("/api/auth/login", "/api/auth/login-code",
                 "/api/health").permitAll()
.requestMatchers("/api/admin/**").hasRole("ADMIN")
.requestMatchers("/api/faculty/**").hasAnyRole("FACULTY", "ADMIN")
.requestMatchers("/api/student/**").hasRole("STUDENT")
.anyRequest().authenticated()
```

Ownership check – AttendanceService.java (Back End)

```
boolean teaches = offering.getFaculty() != null
        && offering.getFaculty().getId().equals(faculty.getId());
boolean coordinates = offering.getSection().getCoordinator() != null
        &&
offering.getSection().getCoordinator().getId().equals(faculty.getId());
if (!teaches && !coordinates && !isAdmin)
    throw new ResponseStatusException(HttpStatus.FORBIDDEN,
            "You are not assigned to this class");
```

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